



BANDARI MARITIME ACADEMY

COMPETENCY-BASED CURRICULUM

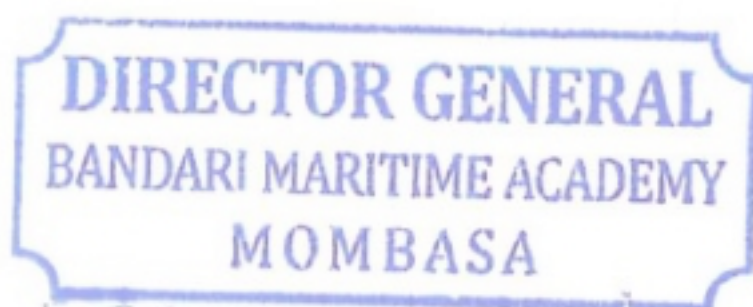
FOR

NAUTICAL SCIENCE TECHNOLOGY

KNQF LEVEL 6

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

PROGRAMME CODE-1041 554 A



First published 2024

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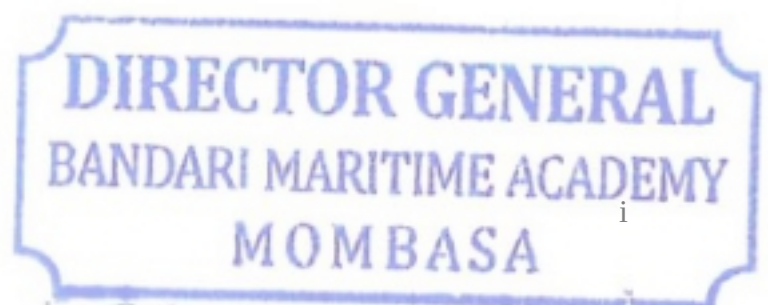
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MOMBASA

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FOREWORD

It is with great pleasure and a sense of responsibility that I introduce the Competency-Based Curriculum (CBC) for the Bandari Maritime Academy. In an era where our global economy is increasingly reliant on maritime trade and transportation, the significance of a competent workforce in the maritime sector cannot be overstated. This curriculum represents a vital step forward in ensuring that our workforce is equipped with the necessary skills and knowledge to navigate the complexities of the maritime industry effectively.

The development of this CBC is a testament to our commitment to fostering a highly skilled workforce that can meet the challenges and demands of the maritime sector in the 21st century.

One of the key features of this curriculum is its competency-based approach. By focusing on the specific skills and abilities required for success in various roles within the maritime industry, this curriculum ensures that learning is practical, relevant, and aligned with industry standards. Whether it is in ship operations, port management, maritime law, or marine engineering, the CBC provides learners with a comprehensive framework for acquiring the competencies needed to excel in their chosen fields.

Moreover, this curriculum is designed to be flexible and adaptable to the evolving needs of the maritime industry. As technology advances and new challenges emerge, it is essential that our workforce remains agile and capable of responding to change. The CBC facilitates lifelong learning and professional development, enabling individuals to stay abreast of industry trends and advancements throughout their careers.

I would like to express my sincere appreciation to all those who contributed to the development of this curriculum, including educators, industry professionals, policymakers, and stakeholders. Your expertise, insights, and dedication have been instrumental in shaping this curriculum into a robust and forward-thinking framework for Nautical Science.

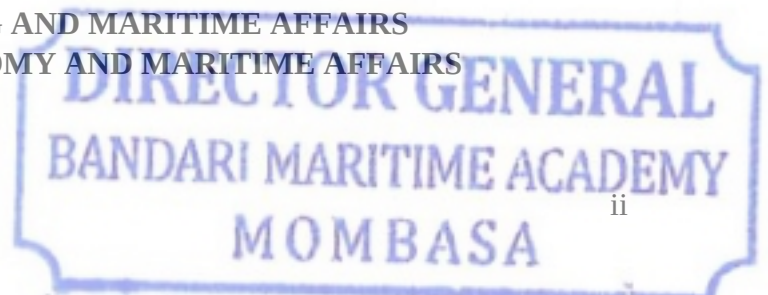
In conclusion, I am confident that the Competency-Based Curriculum for the Bandari Maritime Academy will serve as a cornerstone for excellence in maritime education and training. By equipping our workforce with the requisite skills and competencies, we not only enhance individual opportunities but also strengthen the overall competitiveness and resilience of our maritime industry.



PRINCIPAL SECRETARY

STATE DEPARTMENT FOR SHIPPING AND MARITIME AFFAIRS

MINISTRY OF MINING, BLUE ECONOMY AND MARITIME AFFAIRS



PREFACE

Kenya Vision 2030 aims to transform the country into a newly industrializing, “middle-income country providing a high-quality life to all its citizens by the year 2030”. Kenya intends to create a globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through life-long education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and attitudes necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

The Technical and Vocational Education and Training Act No. 29 of 2013 and Sectional Paper No. 1 of 2019 on Reforming Education and Training in Kenya, emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET in order to address the mismatch between skills acquired through training and skills needed by industry as well as increase the global competitiveness of the Kenyan labour force.

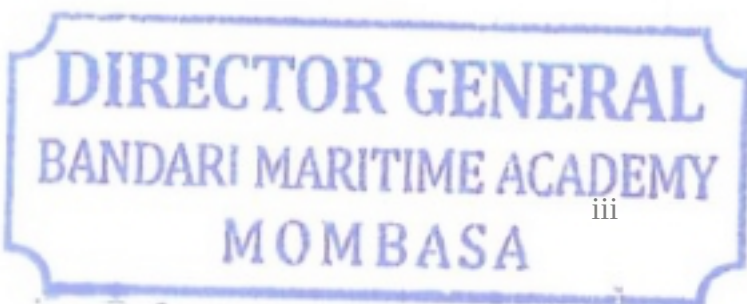
Bandari Maritime Academy in conjunction with, Kenya Maritime Authority, the Nautical Science Sector Skills Advisory Committee (SSAC), Industry representatives, and trainers have developed this curriculum.

This curriculum is designed and organized with an outline of learning outcomes; suggested delivery methods, training/learning resources and methods of assessing the trainee’s achievement. The curriculum is competency-based and allows multiple entry and exit to the course.

I am grateful to the Board members, BMA staff, Kenya Maritime Authority, Nautical Science SSAC, trainers, expert workers and all those who participated in the development of this curriculum.



**CHAIRPERSON
BOARD OF MANAGEMENT
BANDARI MARITIME ACADEMY**



**DIRECTOR GENERAL
BANDARI MARITIME ACADEMY
MOMBASA**

ACKNOWLEDGEMENT

As the Chief Executive Officer of Bandari Maritime Academy, I am honored to extend my heartfelt appreciation and acknowledgment for the development of the Competency-Based Curriculum (CBC).

The introduction of this curriculum marks a significant milestone in the advancement of maritime transport operations training in our nation. It reflects a collaborative effort among educators, industry experts, policymakers, and stakeholders to ensure that our maritime workforce is equipped with the requisite skills and knowledge to thrive in a rapidly evolving industry landscape.

At Bandari Maritime Academy, we are committed to excellence in maritime transport and logistics, and the CBC aligns perfectly with our mission to produce highly skilled professionals who can contribute effectively to the maritime sector. By emphasizing competency-based learning, the curriculum provides our students with a clear roadmap for acquiring the practical skills and expertise needed to excel in their chosen careers.

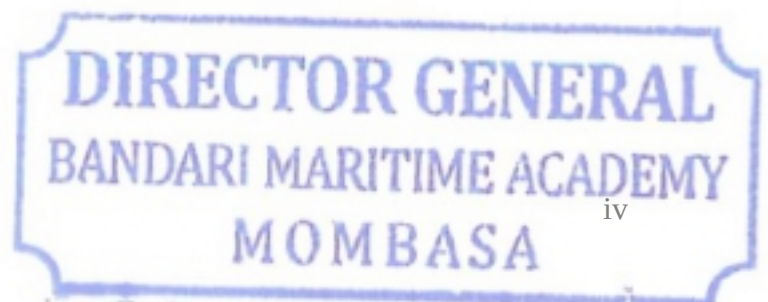
I would like to express my gratitude to all those who played a role in the development of this curriculum, including Kenya Maritime Authority, Nautical Science Sector Skills Advisory Committee (SSAC) members, industry representatives, and trainers for their contribution to the development of this curriculum. Your collective efforts have resulted in a curriculum that is not only comprehensive and relevant but also adaptable to the evolving needs of the maritime industry.

As we embark on the implementation of this curriculum at Bandari Maritime Academy, I am confident that it will serve as a catalyst for innovation and excellence in maritime transport and logistics. By empowering our students with the competencies demanded by the industry, we are not only preparing them for success but also contributing to the growth and prosperity of the maritime sector as a whole.

I extend my best wishes for the successful implementation of the Competency-Based Curriculum in the Academy. Together, let us continue to raise the standards of excellence in maritime transport and logistics, by ensuring a bright and promising future for the maritime industry.



**DIRECTOR/CEO
BANDARI MARITIME ACADEMY**

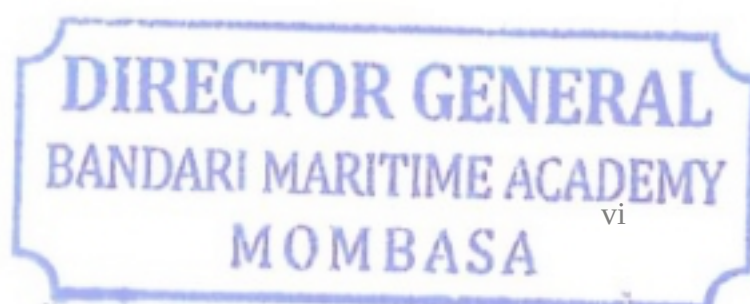


CONTENTS

FOREWORD.....	ii
PREFACE.....	iii
ACKNOWLEDGEMENT.....	iv
COURSE OVERVIEW	vii
BASIC UNITS OF LEARNING.....	x
COMMUNICATION SKILLS	xi
INFORMATION, TECHNOLOGY AND COMMUNICATION	15
ENTREPRENEURSHIP	19
EMPLOYABILITY	23
OCCUPATIONAL SAFETY AND HEALTH PRACTICES	30
COMMON UNITS OF LERANING.....	32
MARITIME COMMUNICATION SKILLS	33
ENGINEERING MATHEMATICS	38
MARINE ENGINEERING PRACTICE	49
MARINE ELECTRICAL AND ELECTRONICS PRINCIPLES	54
COLLISION PREVENTION REGULATIONS.....	58
METEOROLOGY.....	64
MARITIME LAW AND CONVENTIONS.....	71
SHIP CONSTRUCTION PRINCIPLES	85
SHIP STABILITY PRINCIPLES	91
MARINE ENVIRONMENT PROTECTION.....	98
BASIC SAFETY TRAINING	104

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RESEARCH METHODS.....	107
CORE UNITS OF LEARNING.....	111
SEAMANSHIP PRACTICES.....	112
NAVIGATION PRINCIPLES.....	118
CHARTWORK PRACTICES.....	124
CELESTIAL NAVIGATION	130
BRIDGE EQUIPMENT AND SYSTEMS.....	134
ELECTRONIC NAVIGATION EQUIPMENT AND SYSTEMS	140
CARGO HANDLING AND STOWAGE	148
WATCHKEEPING PRACTISE	163
VOYAGE PLANNING AND OCEAN PASSAGE.....	174
SHIP HANDLING	179
EMERGENCIES RESPONSE	184
SHIPPING BUSINESS MANAGEMENT.....	194
ANNEX 1 – STCW TABLE A-II/1.....	204



COURSE OVERVIEW

Description of the course

The Nautical Science qualification KNQF level 6 qualification consists of competencies that a person must achieve to enable him/her to be certified as an Officer in Charge of Navigational Watch (STCW Reg II/1). This qualification pack entails conducting ship passage, maintaining safe navigational watch, performing electronic navigation, implementing navigation emergency measures, performing marine communication, performing ship manoeuvres, managing cargo operation, maintaining ship seaworthiness and managing ship operation.

Basic Units of Learning

Unit Code	Unit Title	Duration in Hours	Credit
0031 541 01A	Communication Skills	50	5
0611 541 02A	Information, Technology and Communication Skills	70	7
0413 541 03A	Entrepreneurship Skills	80	8
0031 541 04A	Employability Skills	50	5
1022 541 05A	Occupational Safety and Health Practices	50	5
Total		300	30

Common Units of Learning

Unit Code	Unit Title	Duration in Hours	Credit
1041 551 06A	Maritime Communication skills	50	5
0541 551 07A	Engineering mathematics	100	10
1041 551 08A	Marine Engineering Practice	80	8
1041 551 09A	Marine Electrical & Electronic Principles	80	8
1041 551 10A	Collision Prevention Regulations	80	8
1041 551 11A	Meteorology	80	8
1041 551 12A	Maritime Law & Conventions	70	7
1041 551 13A	Ship Construction Principles	80	8
1041 551 14A	Ship Stability Principles	90	9
1041 551 15A	Marine Environment Protection	70	7
1041 551 16A	Basic Safety Training	50	5
1041 551 17A	Research Methodology	40	3
Total		870	87

Core Units of Learning

Unit Code	Unit Title	Duration in Hours	Credit
1041 551 12A	Seamanship practices	100	10
	Navigation Principles	100	10
1041 551 14A	Chartwork practice	90	9

1041 551 15A	Celestial navigation	100	10
1041 551 16A	Bridge equipment and systems operation	80	8
1041 551 17A	Electronic navigation equipment Operation	80	8
1041 551 18A	Cargo handling and stowage	90	9
1041 551 19A	Watchkeeping duties	80	8
1041 551 21A	Voyage planning and ocean passage	100	10
1041 551 22A	Ship handling	80	8
1041 551 23A	Response to emergencies	70	7
1041 551 24A	Shipping business management	80	8
1041 551 25A	Trade Project	100	10
	Total	1150	115
1041 551 26A	Industrial Attachment	480	48
Grand Total		2800	280

Course attendance

The trainees MUST to attend a minimum of 80% for each of the units of learning to be eligible to sit for the final end of term/semester assessment.

Entry Requirements

An individual entering this course should have any of the following minimum requirements:

- (i) Kenya Certificate of Secondary Education (K.C.S.E.) with a minimum mean grade of C- (minus)

Or

- (ii) Nautical Science level 5/Craft

And/or

- (iii) Equivalent qualifications as determined by Kenya National Qualifications Authority (KNQA)/Kenya Maritime Authority (KMA)

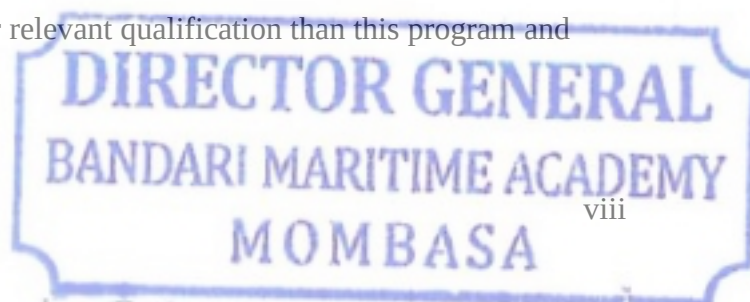
Credit Accumulation, Transfer and exemptions

Bandari Maritime Academy guidelines on credit transfer shall apply.

Trainer qualification

A trainer for this course should have a higher relevant qualification than this program and must have relevant licenses.

Assessment



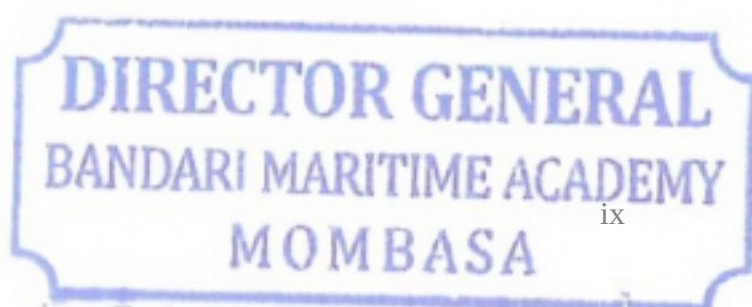
Continuous assessment will be given a weighting of 30% and the end of term/semester assessment supervised by external assessors will be given a weighting of 70% in the determination of the final grade.

RPL Assessment

Assessment for Recognition of Prior Learning (RPL) shall be conducted in line with National RPL requirements and guidelines as well as BMA RPL policy.

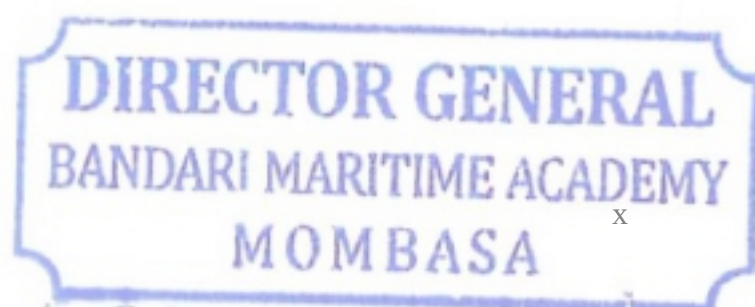
Certification

A candidate will be issued with a **partial certificate of competency** on demonstration of competence in a given unit of core unit of competency. To attain **full certification in Nautical Science level 6** with **respective transcripts**, the candidate must demonstrate competence in all the units of learning as given in this qualification pack. The certificate will be issued by **Bandari Maritime Academy**





BASIC UNITS OF LEARNING



COMMUNICATION SKILLS

UNIT CODE: 0031 541 01A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Demonstrate Communication Skills

DURATION OF UNIT: 50 HOURS

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate communication skills. It involves, meeting communication needs of clients and colleagues; developing communication strategies, establishing and maintaining communication pathways, conducting interviews, facilitating group discussion and representing the organization.

SUMMARY OF LEARNING OUTCOMES

1. Meet communication needs of clients and colleagues
2. Develop communication strategies
3. Establish and maintain communication pathways
4. Promote use of communication strategies
5. Conduct interview
6. Facilitate group discussion
7. Represent the organization

Learning Outcomes, Content, delivery method, and duration

Learning Outcome	Content	Delivery method	Duration (hours)
1. Meet communication needs of clients and colleagues	• Communication process • Modes of communication • Medium of communication • Effective communication • Barriers to communication • Flow of communication • Sources of information	• Discussion • Role playing • Simulation • Direct instruction	10

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Learning Outcome	Content	Delivery method	Duration (hours)
	- Organizational policies - Organization requirements for written and electronic communication methods - Report writing - Effective questioning techniques (clarifying and probing) - Workplace etiquette - Ethical work practices in handling communication - Active listening - Feedback - Interpretation - Flexibility in communication - Types of communication strategies - Elements of communication strategy		
2. Develop communication strategies	- Dynamics of groups - Styles of group leadership - Openness and flexibility in communication - Communication skills relevant to client groups	- Discussion - Role playing - Simulation - Direct instruction	6

Learning Outcome	Content	Delivery method	Duration (hours)
3. Establish and maintain communication pathways	Types of communication pathways	- Discussion - Role playing - Simulation - Direct instruction	4
4. Promote use of communication strategies	- Application of elements of communication strategies - Effective communication techniques	- Discussion - Role playing - Simulation - Direct instruction	6
5. Conduct interview	- Types of interviews - Establishing rapport - Facilitating resolution of issues - Developing action plans	- Discussion - Role playing - Simulation - Direct instruction	6
6. Facilitate group discussion	- Identification of communication needs - Dynamics of groups - Styles of group leadership - Presentation of information - Encouraging group members participation - Evaluating group communication strategies	- Discussion - Role playing - Simulation - Direct instruction	10
7. Represent the organization	- Presentation techniques - Development of a presentation - Multi-media utilization in presentation	- Discussion - Role playing - Simulation - Direct instruction	8

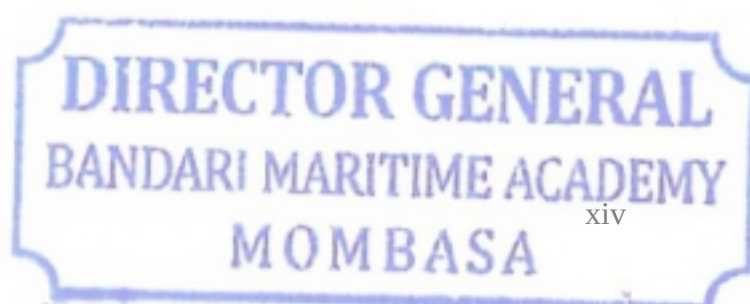
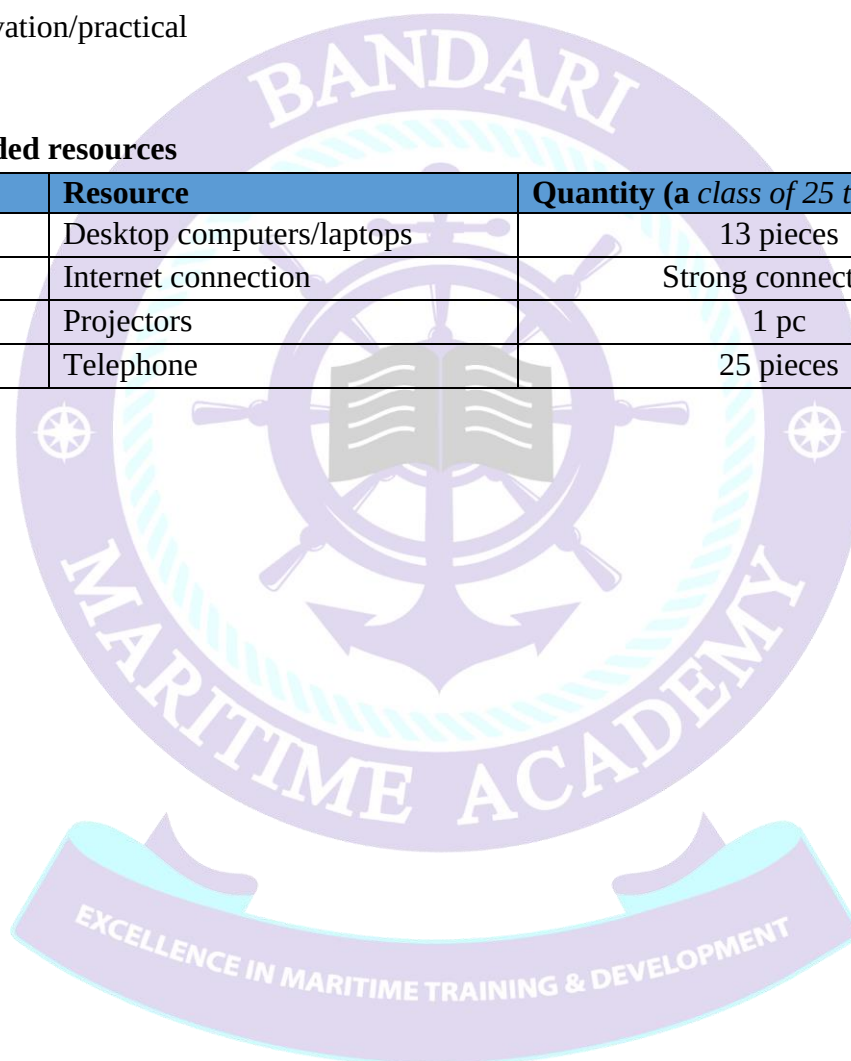
Learning Outcome	Content	Delivery method	Duration (hours)
	Communication skills relevant to client groups		

Suggested methods of assessment

- Oral questioning
- Written assessment
- Observation/practical

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Desktop computers/laptops	13 pieces
2.	Internet connection	Strong connection
3.	Projectors	1 pc
4.	Telephone	25 pieces



INFORMATION, TECHNOLOGY AND COMMUNICATION

UNIT CODE:0611 541 02A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Demonstrate Digital Literacy skills

DURATION OF UNIT: 70 HOURS

UNIT DESCRIPTION

This unit describes the competencies required to demonstrate digital literacy. It involves identifying computer software and hardware, applying security measures to data, hardware, and software in an automated environment, computer software to solve tasks, internet and email in communication at the workplace, desktop publishing in official assignments, and preparing presentations packages.

SUMMARY OF LEARNING OUTCOMES

1. Identify computer software and hardware
2. Apply security measures to data, hardware, and software in an automated environment
3. Apply computer software to solve tasks
4. Apply Internet and email in communication at the workplace
5. Apply desktop publishing in official assignments
6. Prepare presentation packages

Learning Outcomes, Content, delivery method and duration

Learning Outcome	Content	Delivery method	Duration
1. Identify computer hardware and software	• Concepts of ICT • Functions of ICT • History of computers • Components of a computer • Classification of computers	• Instructor-led facilitation of theory • Demonstration by trainer • Practical work by trainee • Viewing of related videos • Group discussions	15
2. Apply security measures to data, hardware, software in	• Data security and control	• Instructor-led facilitation of theory	12

Learning Outcome	Content	Delivery method	Duration
automated environment	- Security threats and control measures - Types of computer crimes - Detection and protection against computer crimes - Laws governing protection of ICT	- Demonstration by trainer - Practical work by trainee - Viewing of related videos - Group discussions	
3. Apply computer software in solving tasks	- Operating system - Word processing - Spread sheets - Data base design and manipulation - Data manipulation, storage and retrieval	- Instructor-led facilitation of theory - Demonstration by trainer - Practical work by trainee - Viewing of related videos - Group discussions	15
4. Apply internet and email in communication at workplace	- Computer networks - Network configurations - Uses of internet - Electronic mail (e-mail) concept	- Instructor-led facilitation of theory - Demonstration by trainer - Practical work by trainee - Viewing of related videos - Group discussions	8
5. Apply desktop publishing in official assignments	- Concept of desktop publishing - Opening publication window	- Instructor-led facilitation of theory - Demonstration by trainer	10

Learning Outcome	Content	Delivery method	Duration
	- Identifying different tools and tool bars - Determining page layout - Opening, saving and closing files - Drawing various shapes using DTP - Using colour pellets to enhance a document - Inserting text frames - Importing and exporting text - Object linking and embedding - Designing of various publications - Printing of various publications	- Practical work by trainee - Viewing of related videos - Group discussions	
6. Prepare presentation packages	- Types of presentation packages - Procedure of creating slides - Formatting slides - Presentation of slides - Procedure for editing objects	- Instructor-led facilitation of theory - Demonstration by trainer - Practical work by trainee - Viewing of related videos - Group discussions	10

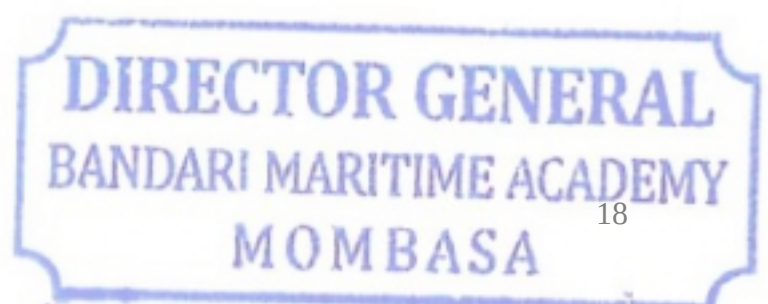
Suggested methods of assessment

- Oral questioning
- Written assessment
- Observation/practical

- Project

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Computers/laptops	13 pcs
2.	Printers	2 pcs
3.	Storage devices	1 pc
4.	Internet access	Strong
5.	Relevant softwares	13 pieces



ENTREPRENEUSHIP

UNIT CODE: 0413 541 03A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the unit of Competency: Demonstrate Entrepreneurial skills

DURATION OF UNIT: 80 HOURS

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate understanding of entrepreneurship. It involves demonstrating understanding of an entrepreneur, entrepreneurship and self-employment. It also involves identifying entrepreneurship opportunities, creating entrepreneurial awareness, applying entrepreneurial motivation and developing business innovative strategies.

SUMMARY OF LEARNING OUTCOMES

1. Demonstrate understanding of who an entrepreneur
2. Demonstrate knowledge of entrepreneurship and self-employment
3. Identify entrepreneurship opportunities
4. Create entrepreneurial awareness
5. Apply entrepreneurial motivation
6. Develop business innovative strategies
7. Develop Business plan

Learning Outcomes, Content, delivery method and duration

Learning Outcome	Content	Delivery method	Duration (hours)
1. Demonstrate knowledge of entrepreneurship and self-employment	• Importance of self-employment • Requirements for entry into self-employment • Role of an Entrepreneur in business • Contributions of Entrepreneurs to National development • Entrepreneurship culture in Kenya • Born or made entrepreneurs	• Direct instruction • Project • Case studies • Field trips • Discussions • Demonstration • Question and answer • Problem solving • Experiential • Team training	10

Learning Outcome	Content	Delivery method	Duration (hours)
2. Identify entrepreneurship opportunities	• Business ideas and opportunities • Sources of business ideas • Business life cycle • Legal aspects of business • Assessment of product demand • Business environment • Factors to consider when evaluating business environment • Technology in business		9
3. Create entrepreneurial awareness	• Forms of businesses • Sources of business finance • Factors in selecting source of business finance • Governing policies on Small Scale Enterprises (SSEs) • Problems of starting and operating SSEs		10
4. Apply entrepreneurial motivation	• Internal and external motivation • Motivational theories • Self-assessment • Entrepreneurial orientation • Effective communications in entrepreneurship • Principles of communication • Entrepreneurial motivation		9
5. Develop business	• Innovation in business • Small business		12

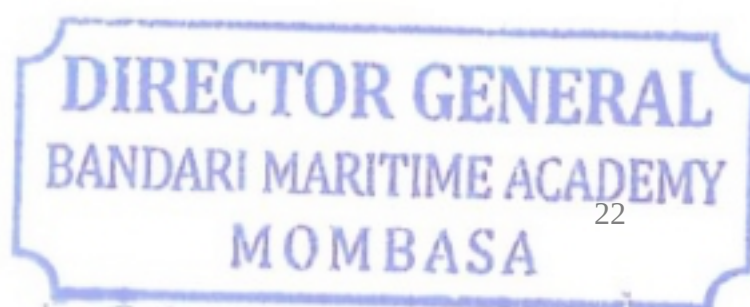
Learning Outcome	Content	Delivery method	Duration (hours)
innovative strategies	Strategic Plan • Creativity in business development • Linkages with other entrepreneurs • ICT in business growth and development		
6. Develop Business Plan	• Business description • Marketing plan • Organizational/Management plan • Production/operation plan • Financial plan • Executive summary • Presentation of Business Plan	•	20
7. Demonstrate knowledge of entrepreneurship and self-employment	• Importance of self-employment • Requirements for entry into self-employment • Role of an Entrepreneur in business • Contributions of Entrepreneurs to National development • Entrepreneurship culture in Kenya • Born or made entrepreneurs		10

Suggested methods of assessment

- Oral questioning
- Written assessment
- Observation/practical
- Individual/group assignments
- Projects
- Third part reports

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Case studies	3
2.	Business plan templates	25 copies
3.	Computers	13 pcs
4.	Overhead projectors	1
5.	Internet	Strong connection
6.	Mobile phone	13 pcs
7.	Video clips	Several
8.	Films	Several
9.	Newspapers and Handouts	25 copies
10.	Business Journals	25 copies



EMPLOYABILITY

UNIT CODE: 0031 541 04A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Demonstrate Employability Skills

DURATION OF UNIT: 50 HOURS

UNIT DESCRIPTION

This unit covers competencies required to demonstrate employability skills. It involves conducting self-management, demonstrating interpersonal communication, critical safe work habits, leading a workplace team, planning and organizing work, maintaining professional growth and development, demonstrating workplace learning, problem solving skills and managing ethical performance.

SUMMARY OF LEARNING OUTCOMES

1. Conduct self-management
2. Demonstrate interpersonal communication
3. Demonstrate critical safe work habits
4. Lead a workplace team
5. Plan and organize work
6. Maintain professional growth and development
7. Demonstrate workplace learning
8. Demonstrate problem solving skills
9. Manage ethical performance

Learning Outcomes, Content, delivery method and duration

Learning Outcome	Content	Delivery method	Duration
1. Conduct self-management	• Self-awareness • Formulating personal vision, mission and goals • Strategies for overcoming life challenges • Managing emotions • Emotional intelligence • Assertiveness versus aggressiveness	• Demonstrations • Simulation/Role play • Group Discussion • Presentations • Question and answer	7

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Learning Outcome	Content	Delivery method	Duration
	- Expressing personal thoughts, feelings and beliefs - Developing and maintaining high self-esteem - Developing and maintaining positive self-image - Setting performance targets - Monitoring and evaluating performance - Articulating ideas and aspirations - Accountability and responsibility - Good work habits - Self-awareness - Values and beliefs - Self-development - Financial literacy - Healthy lifestyle practices - Adopting safety practices		
2. Demonstrate interpersonal communication	- Meaning of interpersonal communication - Listening skills - Types of audience - Public speaking - Writing skills - Negotiation skills - Reading skills - Meaning of empathy - Understanding customers' needs	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	5

Learning Outcome	Content	Delivery method	Duration
	- Establishing communication networks - Assertiveness - Sharing information		
3. Demonstrate critical safe work habits	- Stress and stress management - Time concept - Punctuality and time consciousness - Leisure - Integrating personal objectives into organizational objectives - Resources mobilization - Resources utilization - Setting work priorities - Developing healthy relationships - HIV and AIDS - Drug and substance abuse - Managing emerging issues	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	5
4. Lead a workplace team	- Leadership qualities - Power and authority - Team building - Determination of team roles and objectives - Team parameters and relationships - Individual responsibilities in a team - Forms of communication	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	6

Learning Outcome	Content	Delivery method	Duration
	• Complementing team activities • Gender and gender mainstreaming • Human rights • Developing healthy relationships • Maintaining relationships • Conflicts and conflict resolution • Coaching and mentoring skills		
5. Plan and organize work	• Functions of management • Planning • Organizing • Time management • Decision making concept • Task allocation • Developing work plans • Developing work goals/objectives and deliverables • Monitoring work activities • Evaluating work activities • Resource mobilization • Resource allocation • Resource utilization • Proactive planning • Risk evaluation • Problem solving • Collecting, analysing and organising information	• Demonstrations • Simulation/Role play • Group Discussion • Presentations • Question and answer	6

Learning Outcome	Content	Delivery method	Duration
	Negotiation		
6. Maintain professional growth and development	- Avenues for professional growth - Training and career opportunities - Assessing training needs - Mobilizing training resources - Licenses and certifications for professional growth and development - Pursuing personal and organizational goals - Managing work priorities and commitments - Recognizing career advancement	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	6
7. Demonstrate workplace learning	- Managing own learning - Mentoring - Coaching - Contributing to the learning community at the workplace - Cultural aspects of work - Networking - Variety of learning context - Application of learning - Safe use of technology - Taking initiative/proactivity - Flexibility	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	5

Learning Outcome	Content	Delivery method	Duration
	- Identifying opportunities - Generating new ideas - Workplace innovation - Performance improvement - Managing emerging issues - Future trends and concerns in learning		
8. Demonstrate problem solving skills	- Critical thinking process - Data analysis tools - Decision making - Creative thinking - Development of creative, innovative and practical solutions - Independence in identifying and solving problems - Solving problems in teams - Application of problem-solving strategies - Testing assumptions - Resolving customer concerns	- Demonstrations - Simulation/Role play - Group Discussion - Presentations - Question and answer	6
9. Manage ethical performance	- Meaning of ethics - Ethical perspectives - Principles of ethics - Ethical standards - Organization code of ethics - Common ethical dilemmas	- Demonstrations - Simulation/Role play - Group Discussion - Presentations	4

Learning Outcome	Content	Delivery method	Duration
	• Organization culture • Corruption, bribery and conflict of interest • Privacy and data protection • Diversity, harassment and mutual respect • Financial responsibility/accountability • Etiquette • Personal and professional integrity • Commitment to jurisdictional laws • Emerging issues in ethics	• Question and answer	

Suggested methods of assessment

- Oral questioning
- Written assessment
- Observation/practical
- Portfolio of evidence
- Third part report
- Assignments

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Computers	13 pcs
2.	Stationery	25 pcs
3.	Charts	1 pc
4.	Video clips	Several
5.	Audio tapes	1 pc
6.	Radio sets	1 pc
7.	TV sets	1 pc
8.	LCD projectors	1 pc

OCCUPATIONAL SAFETY AND HEALTH PRACTICES

UNIT CODE: 1022 541 06A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Demonstrate Occupational Safety and Health Practices

DURATION OF UNIT: 50 HOURS

UNIT DESCRIPTION

This unit specifies the competencies required to demonstrate occupational health and safety practices. It involves identifying workplace hazards and risk, identifying and implementing appropriate control measures to hazards and risks and implementing OSH programs, procedures and policies/guidelines.

SUMMARY OF LEARNING OUTCOMES

1. Identify workplace hazards and risk
2. Control OSH hazards
3. Implement OSH programs

Learning Outcomes, Content, delivery method and duration

Learning Outcome	Content	Delivery method	Duration
1. Identify workplace hazards and risks	• Identification of hazards in the workplace and/or the indicators of their presence • Evaluation and/or work environment measurements of OSH hazards/risk existing in the workplace • Gathering of OSH issues and/or concerns	• Discussion • Question & Answer sessions • Roleplay • Viewing of related videos • Industrial visits	15
2. Control OSH hazards	• Prevention and control measures e.g., use of PPE • Risk assessment • Contingency measures	• Discussion • Question & Answer sessions • Roleplay • Viewing of related videos • Industrial visit	15

Learning Outcome	Content	Delivery method	Duration
3. Implement OSH programs	- Company OSH program, evaluation and review - Implementation of OSH programs - Training of team members and advice on OSH standards and procedures - Implementation of procedures for maintaining OSH-related records	- Discussion - Question & Answer sessions - Roleplay - Viewing of related videos - Industrial visits	20

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Standard operating and/or other workplace procedures manuals	25 copies
2.	Specific job procedures manuals	25 copies
3.	Machine/equipment manufacturer's specifications and instructions	25 copies
4.	Personal Protective Equipment - Mask - Face mask/shield - Safety boots - Safety harness - Arm/Hand guard, gloves - Eye protection (goggles, shield) - Hearing protection (ear muffs, ear plugs) - Hair Net/cap/bonnet - Hard hat - Face protection (mask, shield) - Apron/Gown/coverall/jump suit - Anti-static suits - High-visibility reflective vest	@25 pieces



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MARITIME COMMUNICATION SKILLS

UNIT CODE: 1041 551 06A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Apply maritime communication skills

DURATION OF UNIT: 50 HOURS

UNIT DESCRIPTION

This unit covers the competencies required to demonstrate maritime communication skills. It involves, applying marine communication phrases, using English language in written and oral communication, using internal communication system onboard, executing commands in emergency situations, recognizing existence of cultural and religious diversities in the maritime sector and preparing shipboard report.

SUMMARY OF LEARNING OUTCOMES

1. Apply marine communication phrases
2. Use English in written and oral communication
3. Use internal communication systems onboard
4. Execute commands in emergency
5. Recognize the existence of cultural and religious diversities in the maritime sector
6. Prepare shipboard reports.

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply marine communication phrases	● marine communication phrases a. Communication procedures: - i.) Distress - ii.) Search and rescue - iii.) Radio medical advice - iv.) Safety communication - v.) Urgency communication - vi.) Environmental protection - vii.) Meteorological and hydrological condition - viii.) Navigational warnings	● Observation ● Written assessment ● Oral assessment ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	- ix.) Pilotage - x.) Helicopter operations - xi.) Icebreaker operations - xii.) Vessel traffic service (VTS) standard phrases b. Onboard communication phrase: - i.) Standard helm orders - ii.) Standard engine orders - iii.) Pilot on the bridge - iv.) Handing over watch - v.) Safety onboard - vi.) Cargo and cargo handling - vii.) Preparing for sea - viii.) Passenger care • Marine communication challenges a. Language barrier b. Cultural differences c. Noise d. Communication equipment failure e. Attitude f. Lack clarity g. Information overload h. Expectations prejudices i. Email overload • Marine communication channels a. Written communication b. Verbal communication c. Nonverbal communication	
2. Use English in written and oral communication	• Terms and concept used in communication • Communication Process • Classification of communication • Forms of Communication • Channels of Communication • Writing Skills • Summary • Report Writing Skills	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Conducting Meetings and Minute Writing - Interviews	
3. Use internal communication systems onboard	- Types of internal communication systems onboard - GMDSS - Intercom - Public address - Telephones	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
4. Execute commands in emergency situations	- Types of emergencies onboard - Sinking - Man overboard - Fire - Grounding - Foundering - Flooding - Piracy - enclosed space rescue - Evacuation - Collision - Power black out - emergency signals - Continuous ringing of ships bell - One long blast - General alarm- seven short and one long blast - Flare guns - Fire in a drum - Signaling flags - Pyrotechnics - May day call - Three long blasts	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
5. Recognize the existence of cultural and religious	- Elements of interpersonal relationship onboard - Good company policies - Good shipboard management	- Observation - Portfolio of Evidence - Project

Learning Outcome	Content	Suggested Assessment Methods
diversities in the maritime sector	c. Good technical competencies d. Interpersonal relationship of all crew and officers • Effective communication skills a. Listening skills b. People skills c. Emotional intelligence d. Appropriate skill selection e. Ethical communication • communication challenges a. Language barrier b. Cultural differences c. Noise d. Communication equipment failure e. Attitude f. Lack clarity g. Information overload h. Expectations prejudices i. Email overload	• Written assessment • Practical assessment • Oral assessment
6. Prepare shipboard reports	• meaning of report • Types of reports a. Oral reports b. Written c. Management d. Operations procedures e. Production stages f. Maintenance, break down and accident reports g. Internal memos • Role of report in a company • Preparation for report writing a. Audience analysis b. Reading skills c. Data collection d. Data analysis	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Suggested Methods Instruction

- Blended/Hybrid learning
- Case study

- Coaching
- Cooperative learning (Think, pair, share, and Jigsaw)
- Demonstration by trainer
- Flipped classroom
- Group Discussions
- Industrials visits/ Industrial training practical's
- Inquiry-based Learning
- Instructor-led
- Journaling
- Practical Exercise by the trainee(s)
- Reflective discussion
- Roleplaying
- Self-study

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• 1 GMDSS Simulator station • Radar reflector • Portable VHF Radio communication • Shipboard Intercom • PA system • Talkback • Company ISM manual	• 2 Sets International code of signals • 1 Set Flags and shapes • 4 sets Pyrotechnics • 4 Copies of the IMO SMCP Resolution A.918(22) • 4 Copies of the IMO Resolution A.954 23) - Proper Use of VHF Channels at Sea • 4 Copies of SOLAS Chapter V Regulations: Safety of Navigation

ENGINEERING MATHEMATICS

UNIT CODE: 0541 551 07A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the unit of competency: Apply engineering Mathematics

DURATION OF UNIT: 100 HOURS

UNIT DESCRIPTION

This unit describes the competencies required to apply engineering mathematics. It involves applying algebra, applying trigonometry and hyperbolic functions, applying complex numbers, performing coordinate geometry, carrying out binomial expansion, applying calculus, solving ordinary differential equations, carrying out mensuration, performing power series, applying statistics, applying vector theory, applying matrix, solving partial differential equations.

SUMMARY OF LEARNING OUTCOMES

1. Apply algebra
2. Apply trigonometry and hyperbolic functions
3. Apply complex numbers
4. Perform coordinate geometry
5. Carry out binomial expansion
6. Apply calculus
7. Solve ordinary differential equations
8. Carry out mensuration
9. Perform power series
10. Apply statistics and probability
11. Apply vector theory
12. Apply matrix
13. Solve partial differential equations

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Methods of Instruction	Duration
1. Apply algebra	• Calculations involving indices • Laws of indices • Base and index • Conversion of bases • Calculating the indicial equations	• Blended/Hybrid learning • Case study • Coaching • Community of practice (CoP) learning strategy • Cooperative learning (Think,	10 hours

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Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Performance of calculations involving logarithms - Laws of logarithms - Solving logarithmic equations - Use of scientific calculator in solving mathematical problems - Obtainment of solution to system of linear equations involving three unknowns - Performance of calculations involving quadratic equations - Quadratic formula - Null factor law - Reduction of equations - Solution of equations reduced to quadratic form - Performing calculations involving sequence and series - Types of sequence in mathematics - Types of series	- pair, share, and Jigsaw) - Demonstration by trainer - Flipped classroom - Group Discussions - Industrials visits/ Industrial training practical's - Inquiry-based Learning - Instructor-led - Practical Exercise by the trainee(s) - Reflective discussion - Roleplaying - Simulation	
2. Apply trigonometry and hyperbolic functions	- Distinguish trigonometric functions and hyperbolic functions - Definition of trigonometric functions		8 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Properties of trigonometric functions - Evaluation of trigonometric functions - Definition of hyperbolic equations - Properties of hyperbolic functions - Evaluations of hyperbolic functions - Performing calculations using trigonometric rules - Cosine rule - Sine rule - Tan rule - Performing calculations using hyperbolic functions $\sinh x$ $\cosh x$ $\operatorname{cosech} x$ $\coth x$ $\tanh x$ $\operatorname{sech} x$ - The hyperbolic identities		
3. Apply complex numbers	- Representation of complex numbers on the Argand diagram - Definition of complex numbers - The argand diagram		12 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Stating complex numbers in terms of conjugate argument and modulus - Performance of operations involving complex numbers - Addition of complex numbers - Subtraction of complex numbers - Multiplication of complex numbers - Division of complex numbers - Arithmetic operation of complex numbers using De Moivre's theorem - De Moivre's theorem is applied		
4. Perform coordinates geometry	- Calculations involving polar equations coordinate geometry - Drawing of graphs of given polar equations using the Cartesian plane - Determination of normal and tangents using coordinate		7 hours
5. Carry out binomial expansion	- Determination of roots of numbers using binomial theorem - Root extraction		7 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Binomial Theorem is applied in Divisibility and Remainder - Estimation of errors of small changes using binomial theorem		
6. Apply calculus	- Determination of derivatives of functions using differentiation - Differentiation from first principle - Notations - Common derivatives - Rules of differentiation ✓ Product rule ✓ Quotient rule ✓ Chain rule - Determination of derivatives of hyperbolic functions using differentiation - Determination of derivatives of inverse trigonometric functions are determined using Differentiation - Determination of rate of change and small change using differentiation - Performance of calculation involving stationery points of functions of two variables using differentiation - Determination of integrals of algebraic		10 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	functions using integration • Indefinite and definite integral • Methods of integration ✓ Substitution ✓ By parts ✓ Partial fractions • Determination of integrals of trigonometric functions using integration • Determination of integrals of logarithmic functions using integration • Determination of integrals of hyperbolic and inverse functions using integration • Integration is applied		
7. Solve ordinary differential equations	• Solving first order equations using different methods • Separation of variables • Linear equations • Homogenous equations • Solving second order differential equations using the method of undetermined coefficients • The method of undetermined co-efficient • Solving first order and second order differential equations		10 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	from given boundary conditions First order and second order differential equations are applied		
8. Carry out mensuration	- Perimeter and areas of figures are obtained - Perimeter - Area - Obtain volume and surface area of solids - Box ✓ Surface Area of a Rectangular Solid ✓ Volume of a Solid with a Matching Base and Top ✓ Volume of a Rectangular Solid (specific type of solid with matching base and top) - A cylinder ✓ Volume of a cylinder ✓ Surface Area of a Cylinder - Obtain Area of irregular figures using geometry rules - Obtain Areas and volumes using Pappus theorem		6 hours
9. Perform power series	Obtainment of power series using Taylor's Theorem		10 hours

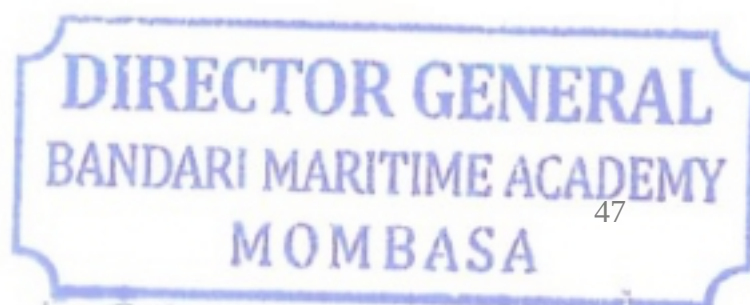
Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Power series using Taylor's theorem - Obtainment of power series using McLaurin's 's theorem - How to find the McLaurin's series - Conditions for McLaurin's series - Taylor's theorem and McLaurin's theorems in numerical work is applied		
10. Apply statistics and probability	- Obtainment of mean, median, mode and standard deviation from given data - Performance of calculations based on laws of probability - Laws of probability ✓ Addition rule ✓ Multiplication Rule - Performance of calculations involving discrete and continuous random variables - Performance of calculation involving probability distributions, mathematical expectation sampling distributions - Binomial distribution		10 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
	- Poisson distribution - Normal distribution - Sampling distribution methods in data analysis is applied - Performance of calculations involving use of standard normal table, sampling distribution, t-distribution and estimation - Sampling distribution - Distribution - Estimation - Determination of confidence intervals		
11. Apply vector theory	- Obtainment vectors and scalar quantities in two and three dimensions - Performing vector operations - Addition - Subtraction - Multiplication - Division - Obtainment position of vectors - Performance of vector resolution - Scalar (Dot) product of vectors		10 hours

Learning Outcome	Content	Suggested Methods of Instruction	Duration
12. Apply matrix	- Obtainment of determinant and inverse of 3x3 matrix - Determinant and inverse of 3x3 matrix are obtained as per the method. - Addition and subtraction of matrices - Determinant of a 3x3 matrix - Inverse of a 3x3 matrix - Obtainment of solutions of simultaneous equations in three unknowns - Calculation involving Eigen values and Eigen vectors		10 hours
13. Solve partial differential equations	- Obtainment of solutions of first order partial differential equations - Obtainment of solutions of second order partial differential equation - The methods of separation of variables - Partial differential equations are applied		10 hours

Suggested methods of assessment

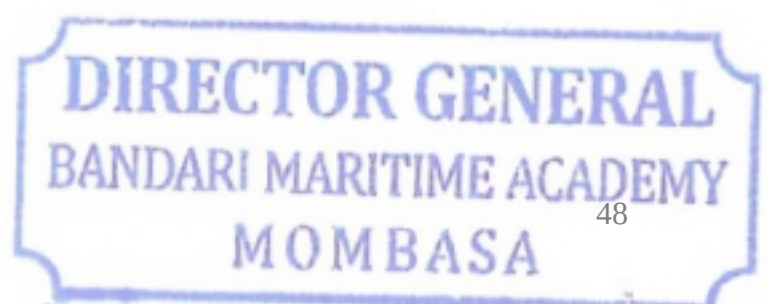
- Oral questioning
- Written assessment



- Observation/practical
- Assignments
- Projects

Recommended resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Mathematical tables	25
2.	Whiteboards	2
3.	Whiteboards markers	6
4.	Scientific calculators	25
5.	Geometrical sets	25
6.	Computers	25
7.	MATLAB software installed in twenty-five (25) Computers	1
8.	Twenty-five (25) rulers	25
9.	Graph books	25
10.	A4 Papers	1 ream
11.	Pencils	25



MARINE ENGINEERING PRACTICE

UNIT CODE: 1041 551 08A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: perform marine engineering practice

DURATION OF UNIT: 80 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to perform marine engineering practice. It involves Operating main machinery, operating auxiliary machinery, operating propulsion control systems, operating bilge, ballast and cargo pumping systems, operating steering gear system, operating electrical, hydraulic and pneumatic systems, operating automated ship control systems (UMS) and maintaining workshop tools and equipment.

SUMMARY OF LEARNING OUTCOMES

1. Operate Main Machinery
2. Operate Auxiliary Machinery
3. Operate Propulsion Control Systems
4. Operate bilge, ballast and cargo pumping systems
5. Operate Steering Gear System
6. Operate Electrical, Hydraulic and Pneumatic Systems
7. Operate Automated Ship Control System (UMS)

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Operate Main Machinery	• Risk assessment for main machinery • Toolbox meeting • PPE • Safety and warning signs around	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of

Learning Outcomes	Content	Suggested Assessment Methods
	the machine's area of operation • Equipment functionality for operational readiness • startup procedure • main machinery operations • shutdown procedure • Records of main Machinery Performance	evidence • Project • Practical
2. Operate Auxiliary Machinery	• Risk assessment for auxiliary machinery • Toolbox meeting • PPE • Safety and warning signs around the machine's area of operation • Equipment functionality for operational readiness • startup procedure • auxiliary machinery operations • shutdown procedure • Records of auxiliary machinery Performance	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Operate Propulsion Control Systems	• engine room standby procedures • pre-departure checklist • Testing Telegraphs engine orders • gangway watchkeepers coordination • Slacking Mooring tension • Testing Forward and astern propulsion • Records of Standby time, to master	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	M/E readiness and completion of pre-departure checklist • Testing Astern propulsion for arrival condition	
4. Operate bilge, ballast and cargo pumping systems	• Risk assessment for bilge, ballast and cargo pumping systems • Toolbox meeting • Firefighting equipment for cargo or fuel tanks • startup procedure for bilge, ballast and cargo pumping systems • Monitoring performance of pump (s) in operation • sounding data from relevant tanks • shutdown procedure • Record of Machinery and systems Performance	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Operate Steering Gear System	• start-up procedure • Steering gear test • Rudder indicator Checks • Records of steering gear test a. Date, b. time and c. position of steering gear	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Operate Electrical, Hydraulic and Pneumatic Systems	• Risk assessment for electrical, Hydraulic and Pneumatic Systems • Toolbox meeting • Stabilizing Electrical, Hydraulic and Pneumatic Systems.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of

Learning Outcomes	Content	Suggested Assessment Methods
	• Guards and protection devices • PPE • Risk of Exposure • Working tools • Safe Machinery operation • Housekeeping	evidence • Project • Practical
7. Operate Automated Ship Control System (UMS)	• Risk assessment for Automated Ship Control System (UMS). • Safe manning document for ship operation • Completion of checklist for approval • Fixing Firefighting appliances to auto mode • bridge duty engineer responsibilities • Dead man alarm • Fire loops with no isolation • Powering down Incinerator and oily water separators • Record of operating Automated ship control system	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies

- Assignments

Recommended Resources for 24 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Computer -Based Training (CBT) modules • Shipboard training • Demonstration equipment • Simulators • Fully equipped Marine engineering workshops	• Convention on the International Regulations for Preventing Collisions at Sea, 1972 • Aldis lamp • Ships whistle • Bell and Gong • Lights and shapes • Compass • Radar and ARPA • Nautical Charts/ECDIS • Company ISM manual

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

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MARINE ELECTRICAL AND ELECTRONICS PRINCIPLES

UNIT CODE: 1041 551 09A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the unit of competency: Apply principles of marine electrotechnology principles

DURATION OF UNIT: 80 HOURS

UNIT DESCRIPTION

This unit describes the competencies required to apply a wide range of marine electrotechnology principles. It involves use of the concept of basic Electrical quantities, use of the concepts of D.C and A.C circuits in electrical installation, use of basic electrical machine, use of power factor in electrical installation, use of earthing in Electrical installations, apply lightning protection measures, apply electromagnetic field theory and apply electrodynamics

SUMMARY OF LEARNING OUTCOMES

1. Use the concept of basic Electrical quantities
2. Use the concepts of D.C and A.C circuits in electrical installation
3. Use of basic electrical machine
4. Use of power factor in electrical installation
5. Use of earthing in Electrical installations
6. Apply lightning protection measures
7. Apply Electromagnetic field theory
8. Apply Electrodynamics

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Use the concept of basic Electrical quantities	• The meaning of SI unit • SI unit of various types of Electrical parameters • Ohm's law • Capacitors and inductors • Calculations involving various Electrical parameters e.g. Power, Current, Voltage, Resistance	• Written tests • Oral questioning • Assignments • Supervised exercises •	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
	Instruments used in measuring various types of Electrical parameters		
2. Use the concepts of D.C and A.C circuits in electrical installation	- Meaning of terms - AC and DC, parallel and series circuits - AC and DC networks - R-L-C transients - AC to DC and DC to AC Conversion - Basic solar photovoltaic systems - Delta star conversion - Definition of power factor - Power triangle - Power factor correction	- Written tests - Oral questioning - Assignments - Supervised exercises	10 hours
3. Use Two Port networks	- Meaning passive networks - Types of Passive network - Characteristic impedance in T & pie networks - Design of T & pie networks - Transmission lines - ABCD Constants - Network in cascade	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours
4. Use of basic electrical machine	- Types of Electrical machines - DC machines, - AC Single and three phase motors, generators and Transformers - Application of AC and DC machines - Special machines and their Applications	- Assignments - Oral questioning - Supervised exercises - Written tests - Practical tests	10 hours
5. Use of earthing in Electrical installations	- Definition of terms in earthing - Earthing points in Electrical installation	- Assignments - Supervised exercises	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
	- Types of earthing systems - Factors to consider in selecting an earthing method - Testing an earthing system	- Written tests - Practical test	
6. Apply electrical system protection	- Electrical faults and abnormal conditions - Components of protection systems - Protective devices - Testing of protection systems - Coordination of protection systems	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours
7. Apply Electromagnetic field Theory	- Introduction to magnetism - Meaning of Electromagnetic Field Theory - Sources of Electromagnetic Fields - Detectors of Electromagnetic radiation - Application of Electromagnetic waves - Electromagnetics Laws - Faraday's Law - Lenz's law - Fleming's Laws - Properties and Effects of Electromagnetic waves - Wave Characteristics and Shielding - Skin Effect - Energy conservation theorem: - Poyntings' Theorem	- Assignments - Oral questioning - Supervised exercises - Written tests	10 hours

Learning Outcome	Content	Suggested delivery methods	Duration
	• Momentum Energy Flow • Electromagnetic Energy flow		
8. Apply Electrodynamics	• Meaning of Electrostatics • Identification of Electrostatic terms and their meaning • Meaning of terms in magnetostatics • Electrodynamics laws	• Assignments • Oral questioning • Supervised exercises • Written tests	10 hours

Suggested Delivery Methods

- Group discussions
- Demonstration by trainer
- Exercises by trainee

Recommended Resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Scientific Calculators	25 pcs
2.	Relevant reference materials	25 copies
3.	Stationeries	25
4.	Electrical workshop	1
5.	Relevant practical materials	25
6.	Dice	25
7.	Computers with internet connection	25

COLLISION PREVENTION REGULATIONS

UNIT CODE: 1041 551 10A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: apply collision prevention regulations

DURATION OF UNIT: 80 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to apply collision prevention regulations. It involves applying general rules for collision prevention, conduction of vessels in any condition of visibility, conduction of vessels in sight of one another, conduction of vessels in restricted visibility, recognizing lights and shapes, recognizing sound and light signals and applying appropriate sound and light signals.

SUMMARY OF LEARNING OUTCOMES

1. Apply general rules for collision prevention
2. Conduct of vessels in any condition of visibility
3. Conduct of vessels in sight of one another
4. Conduct of vessels in restricted visibility
5. Recognize lights and shapes
6. Recognize sound and light signals
7. Apply appropriate sound and light signals

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Apply general rules for collision prevention	• Responsibilities for achieving compliance outcomes for the ship and the roles of other key personnel • Terms used in Collision prevention regulations • Rules of Collision prevention.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence

Learning Outcomes	Content	Suggested Assessment Methods
	Exemption under the International Regulations for Preventing Collisions at Sea 1972(COLREG)	- Project - Practical
2. Conduct of vessels in any condition of visibility	- General applications for conduct of vessels in any condition of visibility - Terms under the conduct of vessels in any condition of visibility. - Conduct of vessel navigating in a narrow channel - Requirements on actions taken to avoid collision in any condition of restricted visibility. - Requirements for a vessel using a traffic separation scheme.	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
3. Conduct of vessels in sight of one another	- Application of the rules applicable to vessels in sight of one another. - Conduct of sailing vessels. - Overtaking vessel - Head-on situation - Conduct of vessels in crossing situations - Actions which may be taken by give-away vessel	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcomes	Content	Suggested Assessment Methods
	- Avoiding action by stand-on vessel - Responsibilities between vessels in sight of one another	
4. Conduct of vessels in restricted visibility	- Application of rules under conduct of vessels in restricted visibility - Requirement on speed on a vessel in restricted visibility. - Regulatory requirements for a vessel which detects by radar alone the presence of another vessel - Close quarter situation in determining risk of Collision - Restrictions of alterations of course in close quarter situations - Action to be taken when a vessel hears any signal forward of the beam	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
5. Recognize lights and shapes	- Application of light and shape rules - Terms on lights and shapes. - Visibility requirements of lights - Lights exhibited by a power-driven vessel underway.	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence

Learning Outcomes	Content	Suggested Assessment Methods
	• Lights exhibited by a towing and pushing vessel • Lights exhibited by a sailing vessels and vessels under oar • Lights exhibited by a fishing vessel • Lights exhibited by vessels not under command or restricted in their ability to manoeuvre • Lights exhibited by a pilot vessel • Lights exhibited by a anchored vessels and vessel aground • Lights exhibited by a sea planes and WIG • Light and shapes positioned and handled	• Project • Practical
6. Recognize sound and light signals	• Terms used for sound and light signals • Equipment for sound signals • Manoeuvring and warning signals.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
7. Apply appropriate sound and light signals	• Sound signals made in restricted visibility • Signals to attract attention	• Observation • Written assessment

Learning Outcomes	Content	Suggested Assessment Methods
	• Distress signals. • Sound signalling appliances	• Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

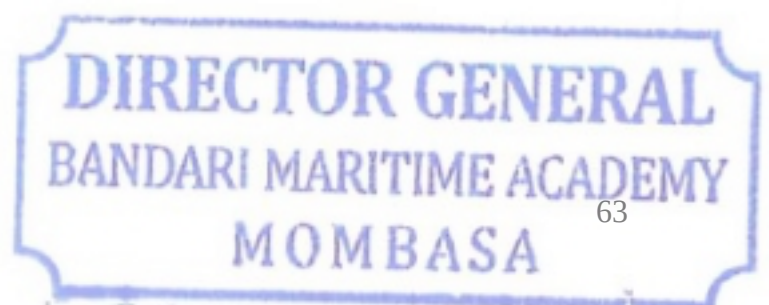
Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector	• Fully equipped seamanship workshop	• Convention on the International Regulations for Preventing Collisions at Sea, 1972 • Aldis lamp • Ships whistle • Bell and Gong • Lights and shapes • Compass • Radar and ARPA

• 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection		• Nautical Charts/ECDIS • Company ISM manual
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METEOROLOGY

ISCED UNIT CODE: 1041 551 11A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Apply Meteorological Practice

DURATION OF UNIT: 80 HOURS

UNIT DESCRIPTION

This unit describes the competencies required by a nautical operations officer to apply meteorological practice. It involves analyzing the features of the major global climate zones, applying knowledge of Winds, applying knowledge of cloud formation and Precipitation, analyzing surface oceanographic processes and data, analyzing the major features of surface synoptic charts, applying meteorological information, using shipboard meteorological instruments, recording weather information, interpreting weather information, applying weather routing in ocean passage planning and monitoring weather conditions.

SUMMARY OF LEARNING OUTCOMES

1. Analyze the features of the major global climate zones;
2. Apply knowledge of Winds;
3. Apply knowledge of cloud formation and Precipitation;
4. Analyze surface oceanographic processes and data;
5. Analyze the major features of surface synoptic charts;
6. Apply meteorological information;
7. Use shipborne meteorological instruments;
8. Record weather information;
9. Interpret weather information;
10. Apply weather routing in ocean passage planning;
11. Monitor weather condition.

Learning Outcome	Content	Suggested Assessment Methods
1. Analyze the features of the major global climate zones	• visibility at sea • Pressure systems, • wind systems, • atmospheric dynamics • Cyclones and Anticyclones	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
2. Apply knowledge of Winds	• Beaufort wind scale • apparent wind and true wind • Combined effects of pressure gradient, Coriolis effect and surface friction • Factors affecting sea surface appearance • The Wind and Pressure systems over the oceans, • Mean Surface Pressure and Wind distribution, • Doldrums, • Inter-tropical Convergence Zones, • Westerlies, • Polar Easterlies, • Monsoons, • Land and Sea Breeze, • Anabatic and Katabatic Winds	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Apply knowledge of cloud formation and Precipitation	• Clouds and precipitation terminology • Main components and cloud formation processes	• Observation • Written assessment • Oral assessment • Third party reports

Learning Outcome	Content	Suggested Assessment Methods
	• Clouds classification • Observation and recording of weather elements • Interpretation and recording of weather elements in navigation practice	• Portfolio of evidence • Project • Practical
4. Analyze surface oceanographic processes and data	• General surface water circulation of the oceans and adjoining seas • Interpretation and recording of ocean current systems/ oceanographic information • Usage of nautical publications on tides and currents • Calculation of tidal conditions. • Forecasting wave height and swell	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Analyze the major features of surface synoptic charts	• Identification of major features of surface synoptic charts • Interpretation and recording of information from synoptic charts • Organisation and updating of shipboard meteorological publications	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Apply meteorological information	• Receiving, recording and interpretation of information on marine weather reports • Identification of weather condition hazards to proposed	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	voyage • Corrective actions taken to avoid weather hazards • Risk mitigation actions implemented to minimise adverse weather effects	• Project • Practical
7. Use shipboard meteorological instruments	• shipboard meteorological instruments • Safe working practices in calibration and use of shipboard meteorological instruments • Safe working practices maintenance of shipboard meteorological instruments • Recording and reporting weather information obtained from shipboard meteorological instruments as per regulatory requirements	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
8. Record weather information	• weather information to be recorded • Meteorological observations, measurements and recording onboard ships • Meteorological reports	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
9. Interpret weather information	• Weather services for shipping industry • Interpretation of weather	• Observation • Written assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	forecasts and oceanographic forecasts Information flow between merchant ships and meteorological stations.	- Third party reports - Portfolio of evidence - Project - Practical
10. Apply weather routing in ocean passage planning	- potential weather hazards to intended passage from weather reports - potential oceanographic hazards to intended passage from oceanographic reports - Weather routing using 'Ocean Passages for the World' and Weather routing nautical charts - Weather risk mitigation through application of company procedures - Dynamic modification of passage plan to address unforeseen meteorological hazards - Documentation of actions taken by the vessel to address meteorological hazards	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
11. Monitor weather conditions	- weather watch - Regular equipment checks for updated weather reports - Regular updating of meteorological records	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	• Passage plan is reviewed, or relevant actions is taken to align with predicted weather conditions • Record observed weather conditions according to workplace requirements. • Disseminate weather information and advice according to workplace procedures.	• Project • Practical

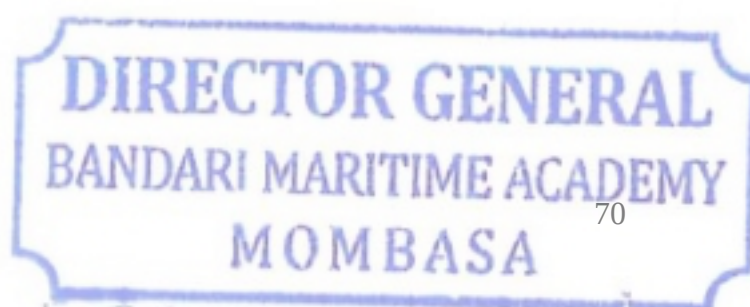
Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software	• Fully equipped seamanship workshop • Wet and dry thermometer	• Weather charts • Company ISM manual

– Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Anemometer • Barometer • Thermometer	
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MARITIME LAW AND CONVENTIONS

UNIT CODE: 1041 551 12A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Monitor Compliance with Legislative Requirements

DURATION OF UNIT: 70 HOURS

UNIT DESCRIPTION

This unit specifies competencies required by an Officer in Charge of a Navigational Watch to monitor compliance with legislative requirements. It involves applying principles of law, applying national maritime legislations, applying safety of life at sea legislation, applying marine environment protection legislation, applying maritime labor legislation, applying maritime training and certification legislation, applying load line legislation, applying maritime security legislation and apply international safety management legislations, applying legislations for carriage of goods and passengers at sea and applying Law of the sea.

SUMMARY OF LEARNING OUTCOMES

1. Apply the principles of Maritime Law
2. Apply National Maritime Legislations
3. Apply Safety of Life at Sea legislation
4. Apply marine environment protection legislation.
5. Apply maritime labour legislation.
6. Apply maritime training and certification legislation.
7. Apply Load line legislation.
8. Apply Maritime Security legislation.
9. Apply international safety management legislations
10. Apply legislations for carriage of goods and passengers at sea
11. Apply Law of the sea

Learning Outcomes, Content and Suggested assessment methods

Learning Outcomes	Content	Suggested assessment methods
1. Apply the principles of Maritime Law	• Meaning of law • Evolution of maritime law • Sources of maritime law • Private and public international law • Introduction to maritime conventions • Process of ratification of international maritime conventions • International maritime organizations a. International Maritime Organization (IMO) b. International Labour Organization (ILO) c. International Telecommunications Union (ITU) • Domestication of international maritime conventions • Mandatory IMO instruments related to safety of navigation (SOLAS, MARPOL, TONNAGE, LOAD LINES, COLREGS, STCW, MLC) • Keeping track with new developments in Maritime Law	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
2. Apply National Maritime Legislations	• Rule of law • Fundamental rights and freedom • Composition of the judiciary and hierarchy of Kenyan courts • Composition and jurisdiction of administrative tribunal in Kenya • National maritime legislation that apply to ships and its personnel • Organisations responsible for compliance with international maritime legislation • Issuance of certificates and endorsement • Provisions for granting dispensation of certificates • Functions of the port State, flag State and Coastal State in enforcing maritime conventions • Ship registration and licensing • Operations and maintenance in accordance with national legislative requirements. • Renewal and extension of certifications to ensure continued validity.	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
3. Apply Safety of Life at Sea legislation	• Historical development of SOLAS Convention • Terms used in the Convention • General provisions of SOLAS • Harmonized surveys and certification (HSCS) • Surveys to which a passenger ship must be subjected • Responsibility for carrying out surveys a. Nominated surveyors b. Recognised Organizations • Certificates issued after survey to ships satisfying the requirements of SOLAS and validity of each of the certificates • Monitoring of shipboard operations in compliance with company operational procedures and regulatory requirements. • Application of the 'no more favorable treatment' clause in accordance with the SOLAS Convention • Provisions for carriage of dangerous goods in bulk in solid form	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
	• Provisions for carriage of dangerous goods bulk in liquid form • Provisions for special requirements for construction of ships and equipment for carriage of packed cargo • Obligations of flag States, port States and coastal States in accordance with SOLAS • Potential non-compliance identified and promptly reported in accordance with Company procedures and legislation • Records prepared and maintained in accordance with regulatory and company requirements.	
4. Apply marine environment protection legislation	• Historical development of MARPOL • Terms used in the MARPOL Convention • General provisions of MARPOL • Regulatory requirements and approvals which apply to ships under MARPOL • Crew awareness of their legislative obligations and	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
	responsibilities with respect to marine environment protection • Monitoring of compliance against company operational procedures and regulatory requirements. • Inspections conducted to monitor compliance against company and regulatory requirements. • Certification requirements under MARPOL • Application of the 'no more favorable treatment' clause in accordance with the MARPOL Convention • Obligations of flag States, port States and coastal States in accordance with MARPOL • Identification of potential non-compliance reporting in accordance with company procedures and legislation • Records prepared and maintained in accordance with regulatory and company requirements.	
5. Apply maritime labour legislation.	• Historical development of Maritime Labour Convention (MLC)	• Oral interview • Project work • Research project

Learning Outcomes	Content	Suggested assessment methods
	• Terms used in the MLC Convention • General provisions of MLC • Regulatory requirements and approvals that apply to ships in accordance with MLC • Rights of seafarers • Obligations of flag States, port States and labour supplying countries in accordance with MLC • Responsibilities of ship-owners based on MLC and national requirements. • Responsibilities of recruitment and placement agents for seafarers based on MLC and national requirements. • Monitoring and compliance of Ships in accordance with Maritime Labour requirements • Application of the 'no more favorable treatment' clause in accordance with the MLC • Identifying potential non-compliance and reporting in accordance with maritime labour regulations • Complaint procedure in accordance with company	• Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
	requirements and regulations Records prepared and in accordance with regulatory and company requirements.	
6. Apply maritime training and certification legislation.	- Historical development of the International Convention on the Standards of Training, Certification and Watchkeeping for Seafarers (STCW) Convention - Terms used in the STCW Convention - Provisions of STCW Convention - Purpose of the Conventions - Maritime Training and certification legislative and regulatory requirements - Control of issuance of certificates and endorsement - Provisions for granting dispensation of certificates - Control functions of the port States in enforcing the provisions of the Convention - Obligations of the flag State maritime training and certification legislation. - Legislative requirements of a ship in accordance with the	- Oral interview - Project work - Research project - Written tests - Individual and group presentations - Case study - Video presentation

Learning Outcomes	Content	Suggested assessment methods
	STCW Convention and national regulatory requirements • Application of the 'no more favorable treatment' clause in accordance with the STCW Convention • Reporting non-compliance in accordance with company procedures and national legislative requirements • Maintain required records in accordance with regulatory and company requirements.	
7. Apply Load line legislation	• Historical development of Load Line Convention, 1966 • Terms applied in the Convention • Load Line regulation a. freeboard b. freeboard deck c. superstructure • Provisions for stowage of deck cargo • Load line convention requirements and approvals that apply to the ship • Certification requirements for certification • Maintenance of records in	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
	accordance with regulatory and company requirements.	
8. Apply Maritime Security legislation.	• Development of the amendment to SOLAS Chapter XI-II • Terms used in ISPS Code • Objectives of the ISPS Code • Responsibilities of contracting Governments under the Code • Concept of declaration of security • Obligations for the ships and company • Control mechanisms provided under the Code • Alternative security arrangements • National regulatory framework for maritime security • Approvals that apply to the ship and port facilities • Obligations of the flag State in accordance with the ISPS Code • Monitoring of compliance under the Code • Identifying potential non-compliance is identified and reporting in accordance with Company procedures. • Certification requirements	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation

Learning Outcomes	Content	Suggested assessment methods
	under the ISPS Code	
9. Apply international safety management legislations	• Historical evolution of the ISM Code • Legislative and regulatory requirements for ships in accordance with the ISM Code a. passenger ships b. tankers and c. bulk carriers of 500gt and upwards • Responsibility of the Master and ship's personnel in accordance with the ISM Code • Inspections are conducted to monitor compliance against company and regulatory requirements. • Opportunities and recommendations for improvements are reported. • Required records are prepared and maintained in accordance with regulatory and company requirements.	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study • Video presentation
10. Apply legislations for carriage of goods and passengers at sea	• Introduction to Athens Convention relating to Carriage of Passengers and their Luggage by sea • Conditions for issuance of Special Passenger Ship Space	• Oral interview • Project work • Research project • Written tests • Individual and group presentations

Learning Outcomes	Content	Suggested assessment methods
	Certificate under the 1973 protocol • Requirements for space for accommodating passengers • Liabilities covered in the PAL Convention • Circumstances for arbitration under the PAL Convention • National legislative and regulatory requirements on the Carriage of goods and passengers at sea • Legislative obligations and responsibilities of crew based on company and regulatory requirements. • Obligations of passengers' in accordance with regulatory based on company and • Required records are prepared and maintained in accordance with company and regulatory requirements	• Case study • Video presentation
11. Apply Law of the sea	• International Conventions on the Law of the Sea a. UNCLOS b. Geneva Convention • Evolution of UNCLOS • Territorial sea • Contiguous zone	• Oral interview • Project work • Research project • Written tests • Individual and group presentations • Case study

Learning Outcomes	Content	Suggested assessment methods
	• Straits used in international navigation • Exclusive economic zone • Continental shelf • High seas • The Area • Concept of innocent passage • Protection of marine environment • Required records are prepared and maintained in accordance with regulatory and organizational requirements.	• Video presentation

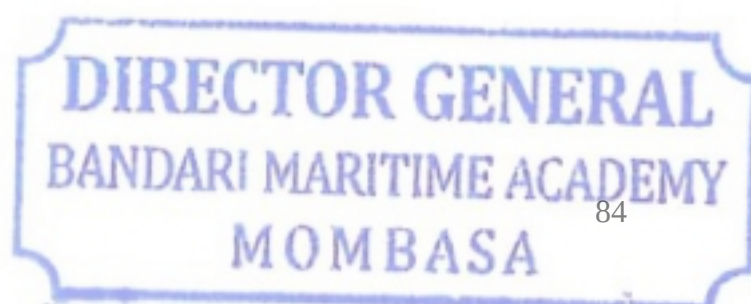
Suggested Methods of Instruction:

1. Group discussions
2. Self-directed learning
3. Demonstrations
4. Role play
5. Practical
6. Project
7. Oral presentations
8. Video presentations

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software	Fully equipped seamanship workshop	• 8 sets of relevant Imo Conventions and Codes

General Resources	Tools and Equipment	Materials and Supplies
c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection		



SHIP CONSTRUCTION PRINCIPLES

UNIT CODE: 1041 551 13A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Apply Ship Construction principles

DURATION OF UNIT: 80 HOURS

UNIT DESCRIPTION

This unit describes the competencies required by a nautical operations officer to identify ship dimension and forms, interpret ship's drawings and plans, identify structural components of a ship, identify structural components of bow and stern, identify ship fittings, identify types of rudders and propeller, identify load lines and draught marks and identify ship stresses.

SUMMARY OF LEARNING OUTCOMES

1. Identify ship dimension and forms;
2. Identify structural components of a ship;
3. Identify structural components of bow and stern;
4. Identify ship fittings;
5. Identify types of rudders and propeller;
6. Identify load lines and draught marks;
7. Identify ship stresses
8. Check weather tight and watertight integrity of ship

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify ship dimension and forms.	• Terms used to describe size and shape of vessels • Terms used to describe key structural components of vessels	• Observation • Project • Written assessment • Oral assessment • Third party report • Interviews

Learning Outcome	Content	Suggested Assessment Methods
	• Terms used to describe size or cargo carrying capacity of vessels • Materials used in hull construction • Properties of high tensile steel • Types of steel used in hull construction • Use of forged, rolled and cast components in hull construction	
2. Identify structural components of a ship.	• Purpose of bulkheads • Midship section of various types of ships • Bulkhead sketched showing construction and attachment to hull • Purpose, construction, and location of collision bulkheads • Test procedures for bulkheads • Maintaining strength in hull openings • Requirements for penetration of collision bulkhead • Longitudinal, transverse, and combined framing	• Observation • Project • Written assessment • Oral assessment • Third party report

Learning Outcome	Content	Suggested Assessment Methods
	- Transition methods from one frame type to another - Maintaining strength of frames when connecting to deck beams and other strength members	
3. Identify structural components of bow and stern	- Panting and pounding forces - Strengthening forepeak sections to resist panting and pounding forces - Anchor and cable arrangements in forepeak tank - Strength members in afterpeak sections - Different rudder support arrangements	- Observation - Project - Written assessment - Oral assessment - Third party report
4. Identify ship fittings.	- Construction and arrangement of ship fittings - Strength members required for deck machinery and fittings - Water tightness and weather tight integrity as required under the SOLAS convention. - Reducing stress a. Hatchways b. Door openings	- Observation - Project - Written assessment - Oral assessment - Third party report

Learning Outcome	Content	Suggested Assessment Methods
	c. Forecastle d. Bridge structure e. Watertight doors f. Gastight doors • Use of suction and discharge valves and fittings in the shell above and below the waterline • Bilge keels structure	
5. Identify types of rudders and propeller	• Propeller Construction • Rudder Construction • Principles of screw propulsion	• Observation • Project • Written assessment • Oral assessment • Third party report
6. Identify load lines and draught marks	• Vessel freeboard • Vessel load line marks • Vessel condition of assignment of freeboard.	• Observation • Written assessment • Project • Oral assessment • Third party report
7. Identify ship stresses.	• Forces acting on a vessel hull • Typical weight and shear force curves • Bending moments • Protection of the hull from corrosion	• Observation • Written assessment • Project • Oral assessment • Third party report
8. Check weather tight and watertight integrity of the ship	• Vessel structural elements critical to weather tightness and watertight integrity	• Observation • Written assessment • Project • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Ship inspections to verify weather tightness and watertight integrity - Ship's load line survey and certification - Ships safety construction survey and certification - Routine checks to confirm weathertight and watertight integrity of a ship - Defects and damage identification, correction and reporting - Effect of severe sea and weather conditions on ship weather tight and watertight integrity	Third party report

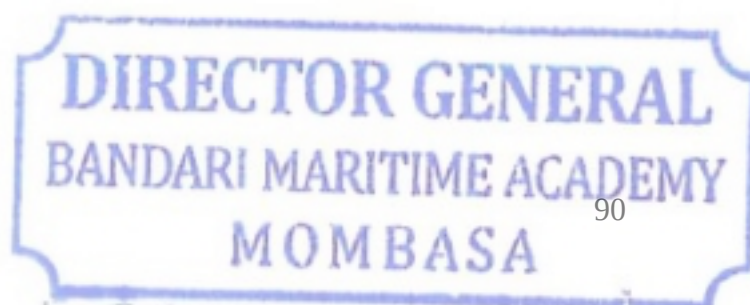
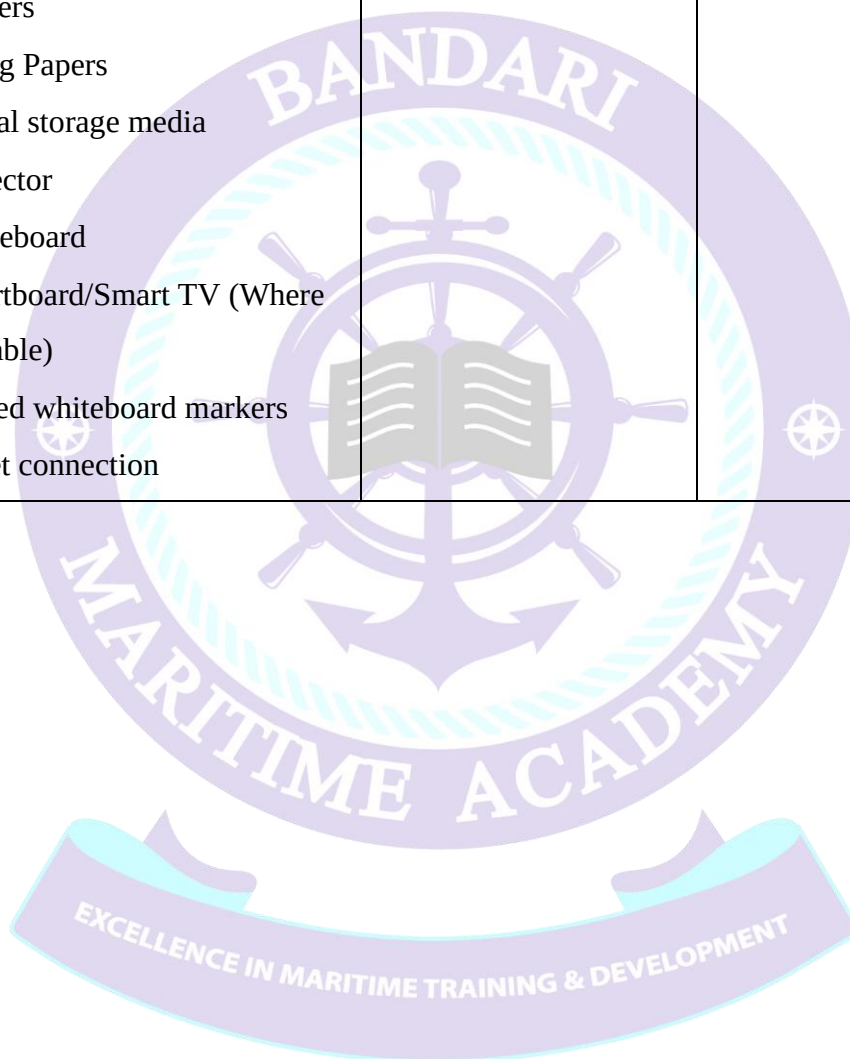
Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Case study

Required resources, tools, equipment and materials (25 trainees)

General Resources	Tools and Equipment	Materials and Supplies
25 computers with the following software:	Fully equipped seamanship workshop	SOLAS Convention

General Resources	Tools and Equipment	Materials and Supplies
a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection		• Ship drawings • Anode bars



SHIP STABILITY PRINCIPLES

UNIT CODE: 1041 551 14A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Apply Ship Stability Principles

DURATION OF UNIT: 90 HOURS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to apply ship stability principles by determining ship displacement and buoyancy, calculating freshwater allowance, determining statical stability, initial stability, angle of loll, curves of statical stability, movement of Centre of gravity, list and its correction and effects of slack tank, calculate trim, determining ship loss of intact stability, buoyancy and ship stresses.

SUMMARY OF LEARNING OUTCOMES

1. Determine ship displacement
2. Determine buoyancy
3. Calculate freshwater allowance
4. Determine statical stability
5. Determine initial stability
6. Determine Angle of Loll
7. Curves of statical stability
8. Determine Movement of Centre of gravity
9. Determine List and its correction
10. Determine Effects of slack tank
11. Calculate Trim and draught
12. Determine ship Loss of intact stability and buoyancy
13. Determine ship stresses stability

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Determine ship displacement	• Forces, Moments and Density – Definitions of Terms – Forces – Mass – Moments • Effects of Density on Stability – Specific Gravity/ Relative Density • Centroids and Center of Gravity – Centroids – COG – Effects of adding mass or Loading Cargo – Effects of Removing Mass or Discharging Cargo – Effects of Shifting Mass or Shifting Cargo – Effects of Suspended Weights • Displacement – Definitions • Lightweight • Deadweight • Displacement • Water Draft • Air Draft – Effect of change of density when the displacement is	• Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	- constant – Effect of density on displacement when the draft is constant • TPC – Definitions • TPC • Water-plane Area – TPC in dock water – TPC Curves • Displacement Curves • Simpson's Rules – Simpson's First Rule – Simpson's Second Rule – Simpson's Third Rule	
2. Determine buoyancy	• Laws of Floatation – Archimedes Principles – Buoyancy – Reserve Buoyancy – Hydrometer • Effect of loss of intact buoyancy • Freeboard	• Written assessment • Oral assessment • Practical assessment
3. Calculate freshwater allowance	• The variable immersion hydrometer • Fresh Water Allowance • Dock Water Allowance	• Written assessment • Oral assessment • Practical assessment
4. Determine statical stability	• Definitions – Heel – List	• Written assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Metacentre • GZ • Moment of Statical Stability • Stiff and Tender Ships • Equilibrium – Stable Equilibrium – Unstable Equilibrium – Neutral Equilibrium	• Practical assessment
5. Determine initial stability	• KB, BM and metacentric diagrams • Righting lever, GZ formula • Relationship between height of metacentre above the keel, KM and the draught • Inclining Experiment	• Written assessment • Oral assessment • Practical assessment
6. Determine angle of Loll;	• Angle of Loll • Capsizing Moments • Correcting Angle of Loll	• Written assessment • Oral assessment • Practical assessment
7. Derive curves of statical stability	• Features of a ship curve of Statical stability • GZ Curves and Stability Criteria • GZ formula for large angles • KN Cross Curves	• Written assessment • Oral assessment • Practical assessment
8. Determine movement of center of gravity	• Movement of Centre of gravity • Effect of load suspension on the ship centre of gravity	• Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• The effect of loading and/or discharging or shifting weights • Change in KG during a passage	
9. Determine List and its correction.	• List • Increase in draft due to list • Correcting List	• Written assessment • Oral assessment • Practical assessment
10. Determine Effects of slack tank	• Free Surface Effect (diagrams/ illustration) • Effects of free surface • Ways to limit Free Surface Effect	• Written assessment • Oral assessment • Practical assessment
11. Calculate Trim and draught	• Longitudinal Stability (diagrams/ illustrations) • Trimming Moments • The moment to change trim one centimeter (MCTC) • Change of draft forward and aft due to change of trim • The effect of shifting weights already on board • The effect of loading and/or discharging weights • Loading a weight to produce a required draft	• Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
12. Determine ship Loss of intact stability and buoyancy	• Effects of loss of stability and buoyancy • Damage Stability • Provisions to prevent adverse effect of loss of intact buoyancy	• Written assessment • Oral assessment • Practical assessment
13. Determine ship stresses	• Ship Stresses • Bending Moments • Shear Forces • Shear forces and bending moments of ships • Weight, buoyancy and load diagrams/ curves	• Written assessment • Oral assessment • Practical assessment

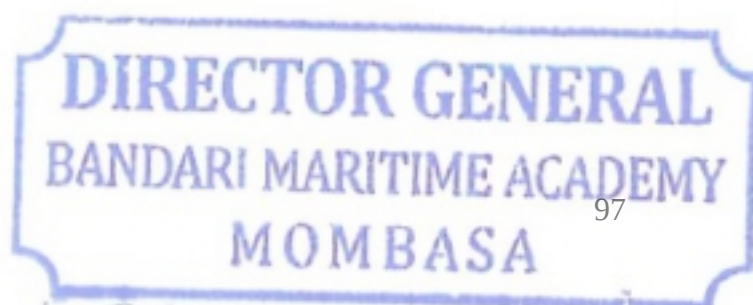
Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Case study

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers	Fully equipped seamanship workshop	• Convention on the International Regulations for Preventing

General Resources	Tools and Equipment	Materials and Supplies
• Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection		Collisions at Sea, 1972 • Aldis lamp • Ships whistle • Bell and Gong • Lights and shapes • Compass • Radar and ARPA • Nautical Charts/ECDIS • Company ISM manual



MARINE ENVIRONMENT PROTECTION

UNIT CODE: 1041 551 15A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Protect Marine Environment

DURATION OF UNIT: 70 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to protect marine environment. It involves applying MARPOL conventions requirements, applying company procedures for the prevention of marine environment pollution, applying the requirements of other conventions for prevention of marine pollution, applying the requirements for the operation of oil pollution prevention, handling hazardous substances safely onboard, applying the requirements of ballast water management and treatment systems, implementing the shipboard oil pollution emergency plans, applying the requirements for volatile organic compound management, operating antifouling systems and maintaining record books, manuals and certificates issued under MARPOL

SUMMARY OF LEARNING OUTCOMES

1. Apply MARPOL conventions requirements on Inspections
2. Apply company procedures for the prevention of marine environment pollution
3. Apply the requirements of other conventions for prevention of marine pollution
4. Apply the requirements for the operation of oil pollution prevention equipment's
5. Handle hazardous substances safely onboard
6. Apply the requirements of ballast water management and treatment systems
7. Apply the requirements for volatile organic compound management
8. Operate antifouling systems
9. Maintain record books, manuals and certificates issued under MARPOL

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Apply MARPOL conventions requirements on Inspections	- Responsibilities for the shipboard personnel and company - compliance monitoring and inspections - monitoring Shipboard operations for compliance - Compliance areas for inspection under port state control regime - Incidents reports.	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
2. Apply company procedures for the prevention of marine environment pollution	- Shipboard procedures for protection of the marine environment. - Company procedures. - Shipboard training and drills - Potential non-compliance - Port State control compliance	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
3. Apply the requirements of other conventions for prevention of marine pollution	- Requirements for other conventions for protection of marine environment - Monitoring Shipboard operations - Incidents reports - Port State Control Compliance - Records	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcomes	Content	Suggested Assessment Methods
4. Apply the requirements for the operation of oil pollution prevention equipment's	• Requirements for shipboard oil pollution prevention equipment • Risk assessment • Safety procedures • Shipboard oil pollution equipment operational procedures • Port State Control Compliance • Operational records	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Handle hazardous substances safely onboard	• Requirements for handling hazardous substances • Risk assessment • Safety procedures for handling hazardous substances • Port State Control Compliance • Records and documentation	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Apply the requirements of ballast water management and treatment systems	• Requirements for operation of ballast water management and treatment system. • Risk assessment • Safety procedures for operation of ballast water management and treatment systems.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	• Ballast water management and treatment systems Operational Procedures • Port State Control Compliance • Management and control requirements • Methods of ballast exchange • Conditions imposed on ballast water exchange • Additional requirements on navigation bridge visibility • Records	
7. Apply the requirements for volatile organic compound management	• Volatile organic compound management systems • compound management. • Sources of VOC onboard ships • Safety procedures for operation of volatile organic compound management systems • Factors that determine VOC emissions • Criteria for control of VOC emissions • Port State Control Compliance • Records	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

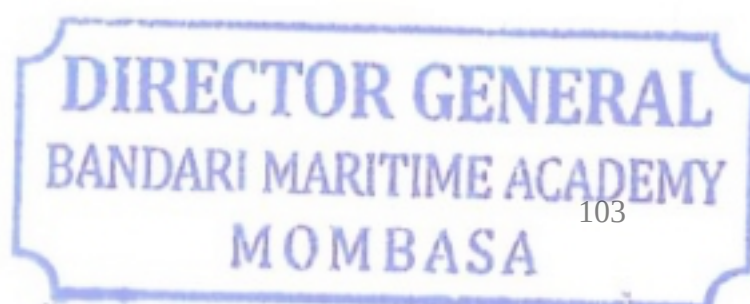
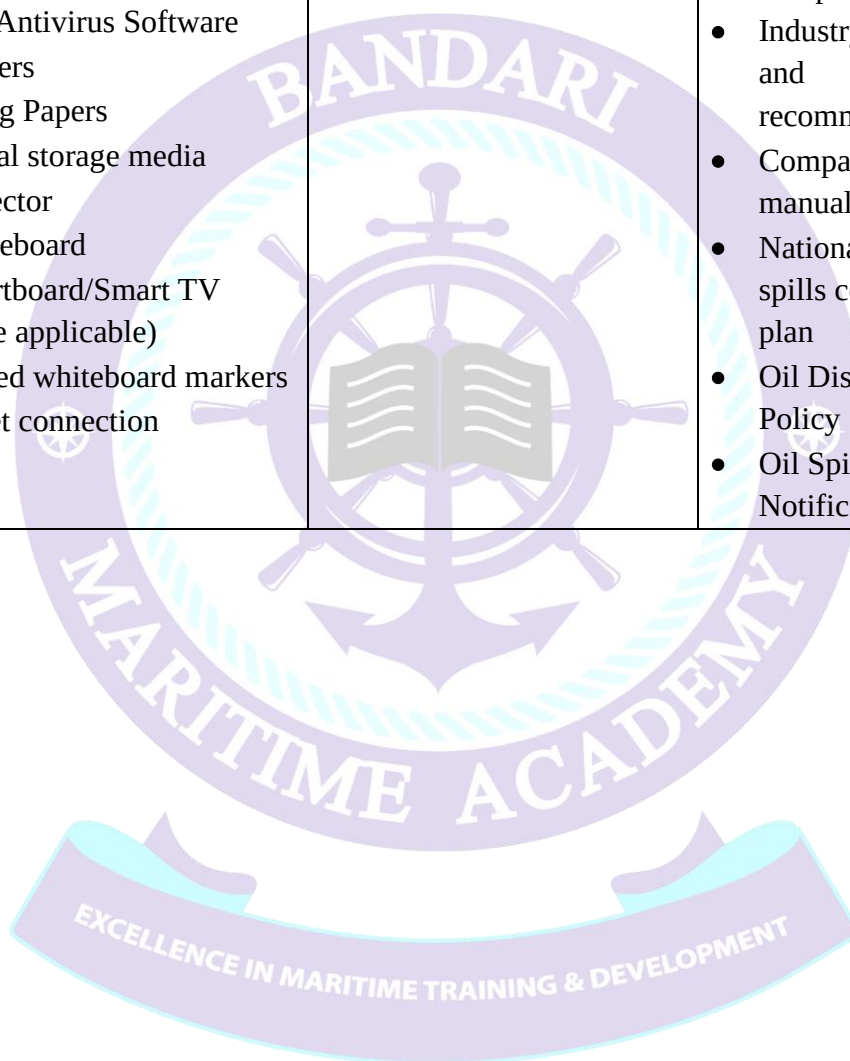
Learning Outcomes	Content	Suggested Assessment Methods
	Requirements for VOC management plan for a tanker carrying crude oil	
8. Operate antifouling systems	- antifouling systems - Safety procedures for operation of antifouling systems - Operational procedures for Antifouling systems - Port State Control and Flag State Compliance - Records and Documentation	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
9. Maintain record books, manuals and certificates issued under MARPOL	- Documentation required and issued under MARPOL convention - Regulatory Compliance - Records and documentation	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 24 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• SOPEP Equipment • ODME - Oil Discharge Monitoring Equipment • Oil Water Separator	• IMO Conventions and codes • National regulations • Merchant shipping notices • Sample SOPEP • Industry circulars and recommendation • Company ISM manual • National marine spills contingency plan • Oil Dispersant Use Policy • Oil Spill Notification form



BASIC SAFETY TRAINING

UNIT CODE: 551 16A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the unit of competency: Demonstrate basic sea survival skills

DURATION OF UNIT: 50 HOURS

UNIT DESCRIPTION

This unit covers the competencies required to effectively demonstrate basic sea survival skills. It entails identifying and being able to utilize lifesaving equipment such as life boat, life jacket, and emergency communication devices such as EPIRB, SART, and being able to swim a minimum of 25 metres.

SUMMARY OF LEARNING OUTCOMES

1. Demonstrate ability to swim
2. Understand the SOLAS convention
3. Identify lifesaving equipment
4. Identify emergency signalling devices
5. Use lifesaving and emergency signalling equipment
6. Apply proper storage and maintenance of life saving equipment

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested delivery methods	Duration
1. Apply first aid techniques	• ABCs of first aid • CPR process • First aid procedures ○ Electric shock ○ Choking ○ Drowning ○ Burns ○ Bone fracture ○ Ingestion of hazardous substances ○ Extreme temperatures	• Practical work by trainee • Demonstration videos • Projects • Group discussions	10 hours
2. Demonstrate ability to swim	• Swimming strokes ○ Freestyle ○ Breast stroke ○ Back stroke • Water treading	• Practical work by trainee • Demonstration videos • Projects	20 hours

Learning Outcome	Content	Suggested delivery methods	Duration
		- Group discussions - Group discussions	
3. Understand the SOLAS Convention	- International Convention for the Safety of Life at Sea (SOLAS) - International Ship and Port Facility Security (ISPS) - Maritime Security (MARSEC) - International Life Saving Appliance Code (LSA)	- Practical work by trainee - Demonstration videos - Projects - Group discussions	5 hours
4. Identify lifesaving equipment	- Lifesaving equipment - Lifesaving crafts	- Practical work by trainee - Demonstration videos - Projects - Group discussions	5 hours
5. Identify emergency signaling devices	Emergency signalling devices	- Practical work by trainee - Demonstration videos - Projects - Group discussions	10 hours
6. Use lifesaving and emergency signaling equipment	- Use of lifesaving equipment - Use of emergency signalling equipment	- Written tests - Practical work by trainee - Demonstration videos - Projects - Group discussions	15 hours

Learning Outcome	Content	Suggested delivery methods	Duration
7. Apply proper storage and maintenance of life saving equipment	- Lifesaving equipment level 1 checks - Lifesaving equipment maintenance	- Practical work by trainee - Demonstration videos - Projects - Group discussions	5 hours

Suggested assessment methods

- Written tests
- Oral
- Practical/Projects
- Demonstration by trainer

Recommended Resources

S/No.	Resource	Quantity (a class of 25 trainees)
1.	Lifesaving equipment	5
2.	Emergency signalling services	5
3.	Communication equipment	5
4.	SOLAS, ISPS, LSA Manuals	25
5.	Stationery	25
6.	First Aid Supplies	2
7.	PPEs	25

EXCELLENCE IN MARITIME TRAINING & DEVELOPMENT

DIRECTOR GENERAL
BANDARI MARITIME ACADEMY
MOMBASA

RESEARCH METHODS

UNIT CODE: 1041 551 17A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the unit of competency: Apply Research methods

DURATION OF UNIT: 40 HOURS

UNIT DESCRIPTION

This unit covers the competencies required to carry out statistical data management. It involves formulating the research problem, carry out literature review, develop research objectives, develop research design and sample design, develop research budget proposal & time plan, collect research data, analyse collected research data, interpret findings and present findings

SUMMARY OF LEARNING OUTCOMES

1. Formulating the Research Problem
2. Carry out Extensive Literature Review
3. Develop research objectives
4. Develop Research Design and Sample Design
5. Develop research budget proposal & Time plan
6. Collecting research Data
7. Analysis of collected research Data
8. Interpretation, Research Findings
9. Present research findings

Learning Outcomes, Content and Suggested Assessment Methods

Learning outcome	Content	Suggested assessment methods
1. Formulating the Research Problem	• Sources of research problems • Definition of research Philosophies • validity and reliability • Characteristics of research • Types of research • The research processes • Sources of research problems	• Written tests • Oral questioning • Assignments • Supervised exercises

Learning outcome	Content	Suggested assessment methods
	Formulation of research problems	
2. Carry out Literature review	- Reviewing the literature - Sources of literature review - Theoretical framework - Conceptual framework - Referencing and citations - Introduction to Google scholar, research gate - Internet search engines	- Written tests - Oral questioning - Assignments - Supervised exercises
3. Develop research objectives/hypothesis or research questions	- Formulation of objectives - main objectives - sub-objectives - characteristics of objectives - characteristics of research hypothesis - formulation of research hypothesis	- Assignments - Oral questioning - Supervised exercises - Written tests
4. Develop Research Design and Sample Design	- Preparing the research design - Identifying Variables - Measurement scales - Study research design - observational - interventional - Types of Sampling techniques - Probability and non-probability - Sample size determination	- Written tests - Oral questioning - Assignments - Supervised exercises
5. Develop research budget proposal & Time plan	- Budget and Costing Development - Direct costs - Indirect costs	- Written test - Observation - Third party report

Learning outcome	Content	Suggested assessment methods
	- Factors to consider when costing - Materials and equipment - Logistics - Administrative - Development of Time plan - Gant charts	- Oral questioning - Interviews
6. Collect research Data	- Methods of Data Collection and their limitations - Research instruments/ data collection tools - Types of questionnaires - Design of questionnaires - Constructing questionnaires - Digitising questionnaires - Mobile technology (ODK) - Piloting the Questionnaire - Ethical issues concerning research participants - Ethical issues relating to the researcher	- Written tests - Oral questioning - Assignments - Supervised exercises
7. Analyse collected research data	- Data Processing - Data management. - Data Analysis Methods	- Written tests - Oral questioning - Assignments - Supervised exercises
8. Interpret research findings	- Interpretations of parameters - Predicting of values	- Written tests - Oral questioning - Assignments - Supervised exercises

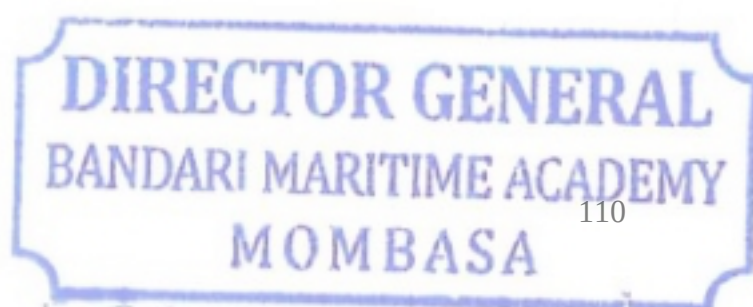
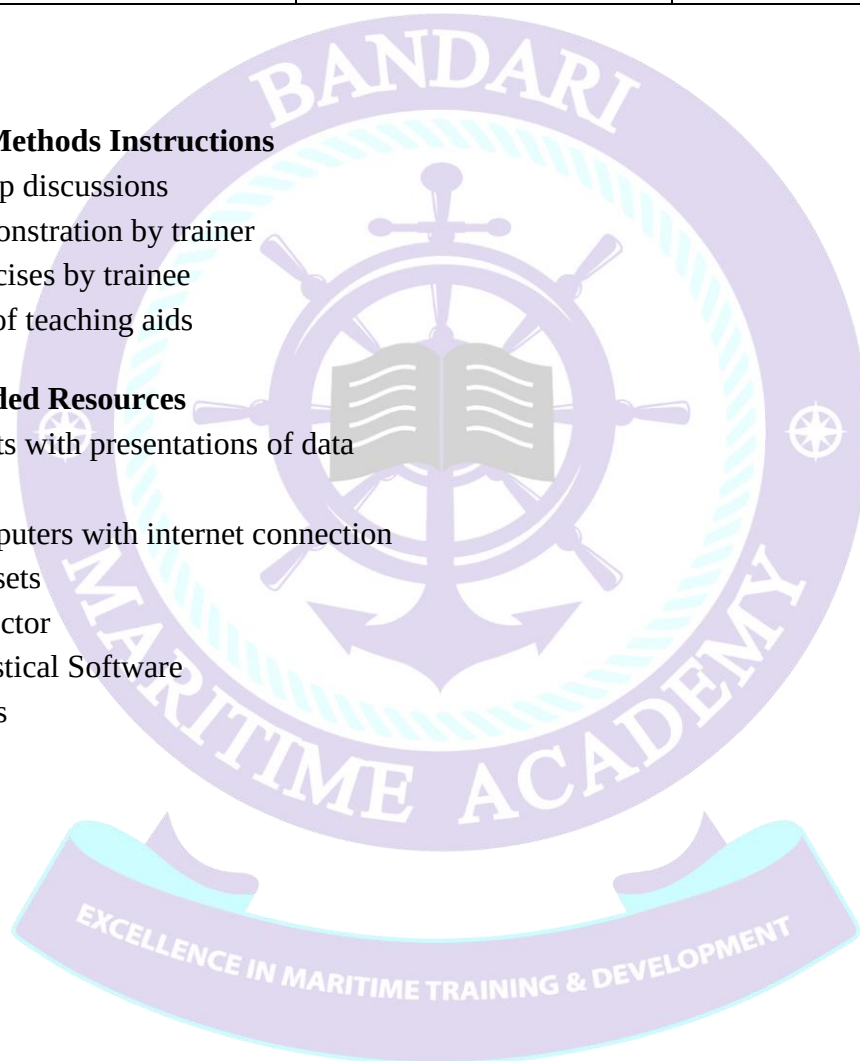
Learning outcome	Content	Suggested assessment methods
9. Present of findings	- reporting of findings - Research Project Report Format - List of References /Bibliography	- Written tests - Oral questioning - Assignments - Supervised exercise

Suggested Methods Instructions

- Group discussions
- Demonstration by trainer
- Exercises by trainee
- Use of teaching aids

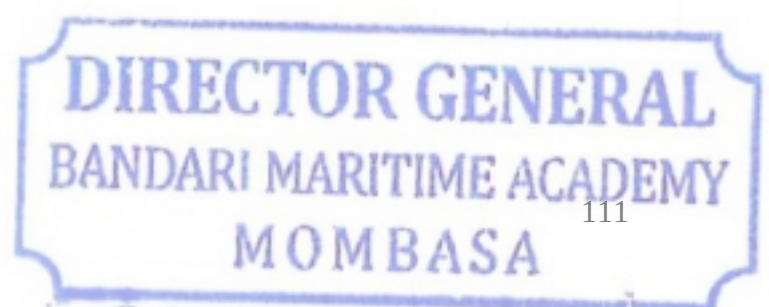
Recommended Resources

- Charts with presentations of data
- Dice
- Computers with internet connection
- Datasets
- Projector
- Statistical Software
- Notes





CORE UNITS OF LEARNING



SEAMANSHIP PRACTICES

UNIT CODE: 1041 551 12A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: perform seamanship practices

DURATION OF UNIT: 150 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to perform seamanship practice. It involves performing anchoring operations, performing mooring and berthing operations, operating deck and cargo handling gear and equipment, performing rope work and rigging, performing ship husbandry, operating Deck machinery and pollution equipment, maintaining cargo handling gear and equipment and also maintaining Deck machinery and pollution equipment.

SUMMARY OF LEARNING OUTCOMES

1. Perform rope work and rigging
2. Perform ship husbandry
3. Perform anchoring operations
4. Perform mooring and berthing operations
5. Operate deck and cargo handling gear and equipment
6. Operate Deck machinery and pollution equipment
7. Maintain cargo handling gear and equipment
8. Maintain Deck machinery and pollution equipment

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Perform rope work and rigging	• Risk assessment for the rigging operation • PPE is worn as per PPE matrix • Ropes for rigging • Rigging shipboard	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project

Learning Outcomes	Content	Suggested Assessment Methods
	equipment • Readiness of the rigged equipment	• Practical
2. Perform ship husbandry	• Risk assessment for maintenance • Work permits • PPE is worn as per PPE matrix • Ship husbandry tools, fittings, equipment • Safety and warning signs • Planned Maintenance System • Housekeeping • Reports for completed maintenance	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Perform anchoring operations	• Anchoring location • Risk assessment • PPE • Proceed to anchor station order • Communication equipment tests • Anchoring equipment tests • Preparations for letting go/heaving up anchor • Anchor operations • The anchor securing • Reporting the position to VTS	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	• Shapes and lights	
4. Perform mooring and berthing operations	• Risk assessment • Toolbox meeting • PPE requirements • Communication equipment tests • Mooring equipment tests • Mooring ropes/lines • Operations associated with Mooring ropes/lines • Making the ship is made fast/ secured • Rat guards	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Operate deck and cargo handling gear and equipment	• Risk assessment for the handling gear and equipment • Toolbox meeting • PPE • Wires, blocks, sheaves inspections • Lashings • Limit switch Tests • Communication equipment tests • The deck and cargo handling gear and equipment operations • Monitoring Performance and efficiency of cargo handling gear and	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	equipment - malfunctions, faults, irregular performance or damage to equipment reports - Equipment shut down and securing	
6. Operate Deck machinery and pollution equipment	- PPE requirements - Communication tests - Deck machinery startup procedures - Deck machinery operational procedures - Pollution equipment set up - Pollution equipment operations - Monitoring Performance and efficiency of equipment - malfunctions, faults, irregular performance or damage to equipment reports - Deck machinery or pollution equipment shut down and securing	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
7. Maintain cargo handling gear and equipment	- Risk assessment - Work permits - PPE requirements - Cargo handling gear and	- Observation - Written assessment - Oral assessment - Third party reports

Learning Outcomes	Content	Suggested Assessment Methods
	equipment inspection. • Tools and parts appropriate for maintenance task • The equipment maintenance • Housekeeping • Maintenance reports	• Portfolio of evidence • Project • Practical
8. Maintain Deck machinery and pollution equipment	• Risk assessment • Work permits • PPE requirements • The equipment/ machinery inspection. • Tools and parts appropriate for maintenance • Maintenance of the equipment • Housekeeping • Maintenance reports	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Fully equipped seamanship workshop • Steering console/navigation simulator • Anchor and anchor chain • Deck machinery • Cargo handling equipment	• Seamanship Techniques by D.J House • Bridge Procedures Guide by International Chamber of Commerce • COSWP Manual • Sample Company ISM Manual • Manila Ropes [various sizes] • 3 sets of synthetic Ropes of various sizes for each trainee • Steel wire Ropes of various sizes for each trainee • Seizing twine and seizing wire • 4 Heaving Lines • 8 Safety Belt • 8 Safety Harness • 8 Marline Spikes • 8 Wooden Spikes

NAVIGATION PRINCIPLES

UNIT CODE: 1041 541 09A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Apply navigation principles

DURATION OF UNIT: 120 hrs

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to apply navigation principles. It involves identifying Earth features for navigations, identifying charts and nautical publications, correcting compass errors, applying parallel and plane sailing, applying Mercator Sailing, applying principles of position fixing, applying tidal principles, and applying great circle sailing principles.

SUMMARY OF LEARNING OUTCOMES

1. Identify Earth features for navigation
2. Identify charts and nautical publications
3. Correct compass errors
4. Apply Parallel and Plane Sailing
5. Apply Mercator Sailing
6. Apply principles of position fixing
7. Apply tidal principles
8. Apply great circle sailing principles

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Identify Earth features for navigation	● Introduction to Navigation ● Definitions ● The Earth ● Rotation ● Revolution ● Meridians ● Parallel of Latitudes ● Latitude and Longitude ● Coordinate System ● Datum	● Written assessment ● Oral assessment ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
2. Identify charts and nautical publications	● Charts Projections – Mercator chart – Gnomonic chart ● Nautical Chart – Chart Information – Chart Scale – Compass Rose – Types of Charts ● Chart Symbols – Depths – TSS – NP 5011 – Np 5012 ● IALA Buoyage System – Lateral and Preferred Channel Marks • Region A • Region B – Cardinal Marks – Safe Water Marks – Isolated Danger Mark – Special Marks – Emergency Wreck Buoys ● IMO Publication Carriage Requirements On Board Vessels – Chart Catalogue – Admiralty Sailing Directions – Admiralty List of Lights – Notices to Mariners – Admiralty Tide Tables – Mariners Handbook – Nautical Almanac – Navigational Tables – Admiralty List of Radio Signals Volumes I to VI – Tidal stream atlases – Ocean Passages for the World – NP 5011	● Written assessment ● Oral assessment ● Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
3. Correct compass errors	• Basic Magnetism • Magnetic and Geographic poles. • Magnetic Variation, Magnetic and True north. • Deviation • Types of Compass • Magnetic Compass – Parts and Function – Types of Compass Cards i. Dry card compass ii. Wet card compass – Deviation Card – Adjusting Magnetic Compass – Maintenance on Magnetic Compass • Gyro Compass	• Written assessment • Oral assessment • Practical assessment
4. Apply Parallel and Plane Sailing	• Definition of Terms – Difference in Latitude – Difference in Longitude – Mean Latitude – Departures – Distance – True Course – Rhumbline • Parallel Sailing ○ Relationship between Difference in Longitude, Latitude and Departure • Plane Sailing • Relationship between Departure, Distance and Course	• Written assessment • Oral assessment • Practical assessment
5. Apply Mercator Sailing	• Mercator Chart Projections • Distance Measuring Between two points on a Mercator Chart • Definition of Terms – Meridional Parts – Mercator bearing • Meridional Parts Table • Relationship between D'long, DMP, Course and Distance	• Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Advantages of Mercator sailing over plane sailing - Vector and Raster Electronic Charts	
6. Apply principles of position fixing	- Position Fixing by GPS - Position Fixing by Compass Bearings. - Position Fixing by Radar Ranges and Bearings - Running Fix - Position Fixing by Horizontal Sextant Angles. - Calculate Distance off by Vertical Sextant Angle. - Position Fixing by Soundings and Range/ Bearing - Calculate the Distance off by Rising/Dipping Distance of Charted Object - ETA - Position Fixing Errors	- Written assessment - Oral assessment - Practical assessment
7. Apply tidal principles	- Definitions of Terms - LAT - MHWS - MLWS - HAT - Set - Rate - Chart Datum - Standard Port - Secondary Port - Causes of Tides - Spring tide - Neap Tide - Relationship between chart datum, LAT, MHWS, MLWS and HAT. - Tide Tables - Tidal Stream Atlas - Tidal diamonds	- Written assessment - Oral assessment - Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Tides and Chart datum • Using the tide tables • Diurnal tides • Semi-diurnal tides • The use of High and Low Water to determine Keel Clearance	
8. Apply great circle sailing principles	• Great Circle Sailing – Definitions a. Great Circle b. Initial Course c. Distances d. Final Course e. Vertex – Feature of Great Circle • Plotting Great circle on Gnomonic and Mercator Charts	• Written assessment • Oral assessment • Practical assessment

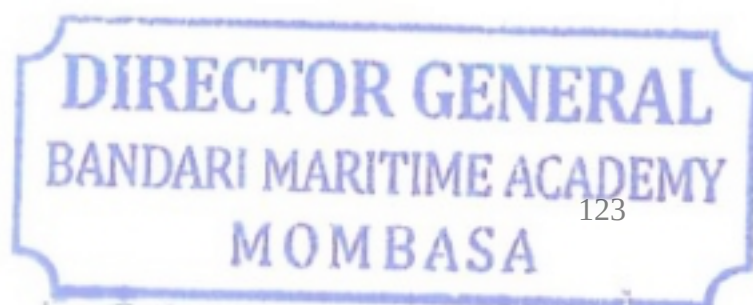
Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable)	Sextant Chronometer Compass Calculator	• Company ISM manual • Plotting pad • Pencil • Pen • Eraser • Stick Glue • Stick • Violet Ink Pens (Approved by UKHO)

General Resources	Tools and Equipment	Materials and Supplies
- Assorted whiteboard markers - Internet connection		- Nautical Charts and Publications - Parallel rule - Compass and dividers



CHARTWORK PRACTICES

UNIT CODE: 1041 551 14A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Perform Chartwork Practices

DURATION OF UNIT: 200 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to perform chartwork practice. It involves selecting Navigation Charts and related Publications, interpreting chart Symbols, updating navigation charts and publications, selecting and update digital charts for navigation, performing tidal calculations, plotting course to steer taking account of the effect of wind/ currents and fixing ship's position using terrestrial objects.

SUMMARY OF LEARNING OUTCOMES

1. Select Navigation Charts and related Publication
2. Interpret chart symbols and IALA Buoyage System
3. Update navigation charts and publication
4. Select and update digital charts for navigation
5. Perform Tidal calculations
6. Plot course to steer
7. Fix Ship's position using terrestrial objects

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Select Navigation Charts and related Publication	• Charts Projections – Mercator chart – Gnomonic chart • Reading Nautical Charts • Chart Scales	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence

Learning Outcome	Content	Suggested Assessment Methods
	• Types of Charts • Tools used in ChartWork • Using Nautical Charts to determine Position, Distance and Bearing • Other Nautical Publications • Information on admiralty charts – Notes – Warnings – Chart Datum – Depth Contours – Landmarks • Chart Accuracy	• Project • Practical
2. Interpret chart symbols and IALA Buoyage System	• Plotting Symbols – Dead Reckoning – Estimated Position – Observed Position – Position Line – Transferred Position – Line – Ground Track – Range/ Distance Off • Heading • Course – Course to Steer – Course Made Good • ETA • Leading Lights and Lines • Bearings	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• Clearing Bearings • IALA Buoyage System – Region A – Region B • Other Chart Symbols (NP5011) • Compass – Basic Magnetsim – Variation and Deviation – Magnetic Compass Error	
3. Update navigation charts and publication	• Admiralty Notice to Mariners – Weekly Notices to Mariners – Cumulative List of Notices to mariners – Annual Summary of Notices to Mariners • Navarea Warnings • Navarea Warnings • Tools used in Chart Correction • Types of Chart Correction – T&Ps – Blocks – Tracings • Manual Chart Correction • The chart correction log	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• Correcting other Admiralty Catalogues	
4. Select and update digital charts for navigation	• ECDIS • ENCs • Chart Correction On ECDIS – Automatically – Manually	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Perform Tidal calculations	• Definitions – LAT – MHWS – MLWS – HAT – Set – Rate – Chart Datum – Standard Port – Secondary Port • Causes of Tides • Spring tide • Neap Tide • Relationship between chart datum, LAT, MHWS, MLWS and HAT. • Tide Tables • Tidal Stream Atlas • Tidal diamonds • Calculate Times and Heights of High Water and	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	Low Water for Standard and Secondary Ports • Co-Tidal Calculations • Calculate and Predict set and rate of tide using tidal diamonds on the chart and tidal stream atlas	
6. Plot course to steer	• Effect of Wind on Speed and Course • Effect of Current on Speed and course • Course to steer allowing for set, drift and leeway • Actual Set and Drift Calculations	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
7. Fix Ship's position using terrestrial objects	• Position Fixing by Compass Bearings. • Position Fixing by Radar Ranges and Bearings • Running Fix • Double angle on the bow • Position Fixing by Horizontal Angles. • Calculate Distance off by Vertical Angle. • Position Fixing by Soundings and Range/ Bearing • Calculate the Distance off by Rising/Dipping Distance of Charted Object	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions

Recommended Resources for 25 Trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: • Windows/Linux/Macintosh Operating System • Microsoft Office Software • Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Sextant • Chronometer • Compass • Calculator	• Company ISM manual • Plotting pad • Pencil • Pen • Eraser • Stick Glue Stick • Violet Ink Pens (Approved by UKHO) • Nautical Charts and Publications • Parallel rule • Compass and dividers

CELESTIAL NAVIGATION

UNIT CODE: 1041 551 15A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Perform Celestial Navigation

DURATION OF UNIT: 150 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to perform celestial navigation. It involves Operating Navigational Equipment for Celestial Navigation, Performing Celestial Observations, Performing Time Calculations, Performing Observer's Latitude Calculations, Calculating Compass Errors and Determining Ship's Position.

SUMMARY OF LEARNING OUTCOMES

1. Operate Navigational Equipment for Celestial Navigation
2. Perform Celestial Observations
3. Perform Time Calculations
4. Perform Observer's Latitude Calculations
5. Calculate Compass Errors
6. Determine Ship's Position

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Operate Navigational Equipment for Celestial Navigation	• Introduction to Celestial Navigation • Composition of the Solar System • Planets used in Celestial Navigation • The Earth	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• Horizon • The Sextant • The Chronometer • Almanac	
2. Perform Celestial Observations	• Concept of Celestial Sphere • Vertical Circles • Observer's Meridian • Cardinal Points • 1st Equatorial System of Coordinates • LHA • Declination • Polar Distance • 2nd Equatorial System of Coordinates • SHA • Right Ascension • GHA • LHA • Bearings • Azimuth • Altitude • Zenith Distance • Twilight a. Civil Twilight b. Nautical Twilight c. Astronomical Twilight	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Perform	• Apparent Solar Day	• Observation

Learning Outcome	Content	Suggested Assessment Methods
Time Calculations	• Mean Solar Day • Sideal Day • Local Mean Time • Greenwich Time • Greenwich Mean Time • Sidereal Time • Calculating GMT of Observations • Relationship between GMT and LMT • The Equation of Time • Zone Time and Zone Number • Dirnual and Annual Motions	• Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Perform Observer's Latitude Calculations	• Application of declination and latitude • Polar distance of lower transit • Calculating Observer's Latitude • Meridian Passage	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Calculate Compass Errors	• Compass Error by Amplitude Method • Compass Error by Time Method • Compass Error by Polaris Method	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Determine Ship's Position	• Simultaneous observations • Plotting Position Lines	• Observation • Written assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Intercept a. Sun b. Star • Plotting Observations	• Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: – Windows/Linux/Macintosh Operating System – Microsoft Office Software – Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection	• Fully equipped seamanship workshop • Sextant • Chronometer	• Compass • Parallel rule • Compass or dividers • Calculator • Plotting pad • Pencil • Pen • Eraser • Navigation charts • Nautical almanac • Norie's Nautical Tables • Star Finder

BRIDGE EQUIPMENT AND SYSTEMS

UNIT CODE: 1041 551 16A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Operate Bridge Equipment and Systems

DURATION OF UNIT: 150 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to operate bridge equipment and systems. It involves; Operating Echo Sounder, Operating GPS, Operating Speed Logs, Operating Voyage Data Recorder (VDR/SVDR), Operating Course Recorder, Operating Automatic Identification System (AIS), Using the Bridge Navigational Watch Alarm System (BNWAS) and Using Long-Range Identification and Tracking (LRIT).

SUMMARY OF LEARNING OUTCOMES

1. Operate Echo Sounder
2. Operate GPS
3. Operate Speed Logs
4. Operate Voyage Data Recorder (VDR/SVDR)
5. Operate Course Recorder
6. Operate Automatic Identification System (AIS)
7. Use the Bridge Navigational Watch Alarm System (BNWAS)
8. Use Long-Range Identification and Tracking (LRIT)
9. Operate Magnetic Compass
10. Operate Gyro Compass
11. Operate Steering Control Systems

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Operate Echo Sounder	• Switching on echo sounder • Display settings • Limitations of Echo Sounder • Alarms and warnings	• Observation • Written assessment • Oral assessment • Third party reports

Learning Outcomes	Content	Suggested Assessment Methods
	• The depth scale of echo sounder • Reading depth displayed	• Portfolio of evidence • Project • Practical
2. Operate GPS	• GPS Switching on Procedures • Display brightness settings • Limitations and errors of GPS • Settings for the GPS for navigation • Ship's position • datum • Alarms and warnings from the GPS	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Operate Speed Logs	• Speed log Switching on Procedures • Display setting • Errors • Speed readings with reference to ground and water	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Operate Voyage Data Recorder (VDR/SVDR)	• VDR/ SVDR switching on Procedures • VDR/SVDR Operational Status • Alarms and warnings from VDR • Information access in case of an incidence	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project

Learning Outcomes	Content	Suggested Assessment Methods
	Information access from the flash drive is read from a computer	Practical
5. Operate Course Recorder	- Removal and loading of printing paper - Checklist for functioning of stylus - Comparing Ship's course to course recorded	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
6. Operate Automatic Identification System (AIS)	- AIS switching on procedures - Display settings - Limitations - AIS set up for navigation - Alarms and warnings - Information displayed on AIS	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
7. Use the Bridge Navigational Watch Alarm System (BNWAS)	- BNWAS switching on procedures - Alarms from BNWAS - BNWAS alarm settings	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcomes	Content	Suggested Assessment Methods
8. Use Long-Range Identification and Tracking (LRIT)	• LRIT switching on procedures • Operation checks for LRIT • Malfunction reports	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
9. Operate Magnetic Compass	• Magnetic compass errors • Topping up Pure ethanol • Funnel effect • Heeling error • Hard iron magnetism • deviation curve • Compass error	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
10. Operate Gyro Compass	• Gyro switching on procedures • Monitoring Gyro compass for stabilization • Aligning Mother gyro with repeaters • Monitoring Gyro compass for inference • Gyro error	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
11. Operate Steering Control Systems	• Steering pumps • Steering pumps checks	• Observation • Written assessment • Oral assessment • Third party reports

Learning Outcomes	Content	Suggested Assessment Methods
	- Steering control (rudder indicator and rudder movement) tests - Change over to hand steering tests - Autopilot settings tests - Change over to non-follow up tests - Solenoids tests - Recording of Test in the movement book	- Portfolio of evidence - Project - Practical

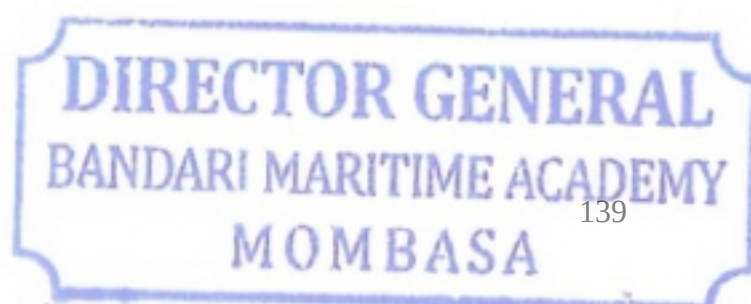
Suggested Methods of Instruction:

- Workshops
- Computer -Based Training (CBT) modules
- Shipboard training
- Demonstration equipment
- Simulators

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
24 computers with the following software: - Windows/Linux/Macintosh Operating System - Microsoft Office Software - Antivirus Software 2 Printers	2 Echo sounder 2 Course recorder 2 LRIT 2 AIS 2 GPS/DGPS 2 VDR/SVDR	- Operating manuals - Company ISM manual

• Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• 2 Speed logs • 2 BNWAS • Access to bridge simulator.	
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ELECTRONIC NAVIGATION EQUIPMENT AND SYSTEMS

UNIT CODE: 1041 551 17A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Operate Electronic Navigation Equipment

DURATION OF UNIT: 120 HRS

UNIT DESCRIPTION

This unit specifies competencies required by a nautical operations officer to operate electronic navigation equipment. It involves setting up and operating navigational radar, setting up and operating navigational ARPA (Automatic Radar Plotting Aid), performing Manual Radar Plotting, operating ECDIS, operating an INS/IBS system and electronic systems of position fixing.

SUMMARY OF LEARNING OUTCOMES

1. Set up and operate navigational Radar
2. Set up and operate navigational ARPA (Automatic Radar Plotting Aid)
3. Perform Manual Radar Plotting
4. Operate an ECDIS
5. Operate INS/IBS system

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Set up and operate navigational Radar	• Carriage requirement and performance • Terms related to radar • Radar set up and shut down procedure • Radar screen settings	• Observation • Written assessment • Oral assessment • Third party reports

Learning Outcomes	Content	Suggested Assessment Methods
	a. Screen brilliance b. Panel brilliance c. Day and night • Radar display mode a. Relative b. True • orientation modes a. North-up b. Head-up c. Course up • Radar Range scale function • Gyro error and accuracy of the heading marker alignment is checked as per master's instructions, equipment manual and company policy • Radar limitations • Radar's performance • Parallel index • Warnings and alarms • Radar tools for position fixing a. Variable Range Marker b. Electronic Bearing Line	• Portfolio of evidence • Project • Practical
2. Set up and operate navigational ARPA (Automatic Radar Plotting Aid)	• Speed input a. Ground stabilised b. Sea stabilised • Vectors a. Relative vectors	• Observation • Written assessment • Oral assessment • Third party reports

Learning Outcomes	Content	Suggested Assessment Methods
	b. True vectors • Target trails a. Relative trails b. True trails • Past position dots switched on as per equipment manual and company policy • Alarm limits a. CPA- Closest Point of Approach b. TCPA-Time to Closest Point of approach • Bow crossing range • Bow crossing time • Guard zones, automatic target acquisition zones and their sensitivity • Target identification and automatic dropping of target • Cursor display information • Integral test programmes • ARPA trial function	• Portfolio of evidence • Project • Practical
3. Perform Manual Radar Plotting	• Target information a. Range b. Bearing • Target track (OA), Own ship track (OW) WA • TCPA (time to closest point of approach) and CPA (closest Point of Approach)	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project

Learning Outcomes	Content	Suggested Assessment Methods
	- Target analysis - Course - Speed - CPA - TCPA - Aspect	Practical
4. Operate an ECDIS	- Terms and definitions used in ECDIS - Types of ECS - Vector - Raster - ECDIS operational setup - Primary and back up ECDIS - ECDIS IMO Performance Standards: - S-52 ECDIS Display and Presentation - S-57 Electronic Navigational Chart (ENC) - S-58 IHO Recommended ENC Validity Checks - S-61 Raster Navigational Chart (RNC) - S-62 ENC Producer Codes - S-63 ENC Data Protection	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcomes	Content	Suggested Assessment Methods
	g. S-64 IHO Test Data Sets for ECDIS h. S-65 ENC Production Guidance • Safety value available in ECDIS • Automatic and manual functions of ECDIS • Voyage recording • Over-reliance and complacency on ECDIS	
5. Operate INS/IBS system	• bridge equipment setup and operational status • Alarms and warnings • Limitations of INS/IBS system	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Operate Electronic systems of Position fixing	• LORAN C and Enhanced LORAN (eLORAN) • Global Positioning System (GPS) and Differential GPS (DGPS) a. GPS system configuration ▪ GPS frequencies ▪ C/A and P codes	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	b. Basic Line measurement c. Dilution of Precision (DOP) d. Errors of GPS e. Accuracy of GPS f. Datum shift g. WGS 84 h. DGPS ▪ Transmission of corrections by DGPS stations ▪ Limitations of the DGPS receiver • Global Navigation Satellite System (GLONASS) a. GLONASS and GPS satellite configuration b. Combined GPS/GLONASS receiver setp and configuration c. limitation of the GLONASS system receiver • GALILEO a. Galileo Satellites and Orbits b. Satellite Geometry and Dual Atomic Clocks in the Galileo System	

Learning Outcomes	Content	Suggested Assessment Methods
	c. Calculating the position of the receiver using atomic clock signal information d. Triangulation of signals from multiple satellite e. Limitations of the Galileo System Receiver • BEIDOU a. Conception and development b. operating Principles c. Frequencies d. Operation e. Accuracy f. Limitations	

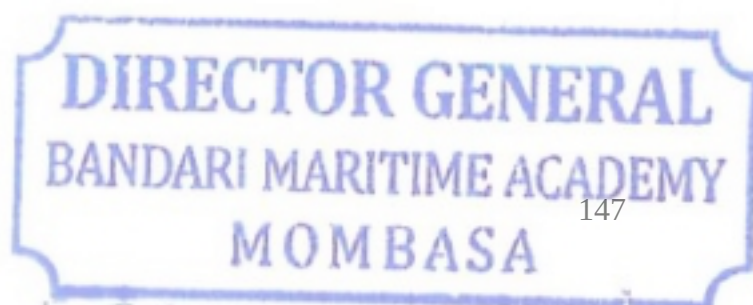
Suggested methods of instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software	• 2 ECDIS • 2 Radar X-Band • 2 Radar S-Band • 2ARPA	• ENC's • Operating manuals • Company ISM manual

General Resources	Tools and Equipment	Materials and Supplies
• 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• 2 Compass • 2 Global Positioning System (GPS) • 2 Differential GPS (DGPS) • Access to a bridge simulator	



CARGO HANDLING AND STOWAGE

UNIT CODE: 1041 551 18A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Perform Cargo Handling and Stowage

DURATION OF UNIT: 150 HRS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to perform cargo handling and stowage. It involves determining ship displacement, determining buoyancy, calculating freshwater allowance, determining statical stability, manage partial loss of intact stability and buoyancy, determining initial stability, deriving curves of statical stability, determining movement of center of gravity, determining effects of slack tank, calculate trim, determining angle of loll, list and its correction, performing stability calculations for bulk cargo, determining loss of intact buoyancy, maintaining records of stability management and determining ship stresses and effect of cargo on stability.

SUMMARY OF LEARNING OUTCOMES

1. Handle Liquid, Chemical and Liquefied Gas
2. Handling unitized cargo
3. Handle Ro-Ro Cargo
4. Handle Ores carried in bulk
5. Handle Heavy Lift Cargo
6. Handle Coal Cargoes
7. Handle Grain Cargo
8. Handle Dangerous Goods (DG) cargo
9. Handle timber deck cargo
10. Perform passenger vessel operations
11. Inspect and Record Cargo Damages and Defects

Learning Outcomes, Content, and Suggested Assessment Methods

Learning outcomes	content	Suggested assessment methods
1. Handle Liquid, Chemical and Liquefied Gas	• Cargo information • Cargo documentation • Cargo handling equipment • Pre-arrival information	• Observation • Written assessment • Oral assessment • Third party reports

Learning outcomes	content	Suggested assessment methods
	a. FAL Convention b. ISGOTT checklist • Pre-arrival checks on cargo tanks and vent systems with/without inert gas system • Safe Vessel moored and berthing • Pre-transfer conference • Pre-transfer checks • Loading operation • Measurement and recording of Temperatures in heated cargo tanks • Inert gas system • Tank gauging equipment • Discharge plan • Inspection and calibration of tank gauging equipment • Pre-arrival checklist • Discharging Operations • Cargo discharge completion checklist	• Portfolio of evidence • Project • Practical
2. Handle unitized cargo	• Cargo information • Cargo documentation	• Observation • Written assessment

Learning outcomes	content	Suggested assessment methods
	on board • Cargo holds and hatch covers preparation • Cargo handling equipment • Cargo calculations • Safe Vessel is moored and berthing • vessel and terminal's representative/planner meeting • Procedures Loading operation • Monitoring Vessel's draft, trim and list during loading • Regular inspections on container/cargo • Loaded DG is regularly sighted and inspected as per its specific IMDG schedule/ risk • Reefer temperature are observed and recorded at regular intervals as per CSC convention and company procedures • Weather conditions • Risk Assessment	• Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	• A tool box meeting • Information is exchanged • Discharging plan • Inspection of Cargo handling equipment • Cargo handling procedures • Discharging operations • Cargo discharge completion checklist	
3. Handle Ro-Ro Cargo	• Cargo documentation • Shipper's information about the cargo • Vessel preparation for receiving cargo • Safe Vessel mooring and safe access • Loading plan meeting. • Procedures for commencement of loading operation • Cargo lashing and the integrity of securing points. • Fire and safety rounds • Information is exchanged with terminal representative at port of discharge	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	- Discharging operation procedures - cargo discharge checklist	
4. Handle Ores carried in bulk	- Cargo documentation - shipper's information about the cargo - Vessel preparation for receiving cargo - Safe Vessel mooring and safe access at port of loading and discharge - Loading plan meeting. - Cargo is inspected prior to loading - Loading operations. Cargo hold monitoring systems. a. Bilge alarms b. Water ingress alarms c. Gas detectors d. Explosimeter e. Thermometers f. Sounding pipes g. Sampling	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning outcomes	content	Suggested assessment methods
	points h. Smoke detectors - Information exchanged with terminal representative at port of discharge, - Discharging operation - Cargo discharge checklist	
5. Handle Heavy Lift Cargo	- Cargo documentation - shipper's information about the cargo - Vessel preparations to receive the heavy lift cargo - Risk assessment and safety meeting - Vessel's stability at the worst condition is calculated and all free surfaces eliminated - Notified of loading operations and hoisting of the flag RY - Inspection and monitoring of Cargo lashing and securing points - Information exchange	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning outcomes	content	Suggested assessment methods
	with terminal representative at port of discharge, including the discharging plan/sequence • Safe Vessel mooring/berthing and safe access to the ship and holds • Discharge operations and procedures • Cargo discharge checklist	
6. Handle Coal Cargoes	• Cargo documentation and shipper's information about the cargo • Vessel preparation to receive Coal cargo • Vessel is safely moored alongside and safe access to the vessel provided as per COSWP • Loading Meeting: a. loading plan, b. loading sequence and c. final stowage Plan	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	• Cargo inspection • Loading operations • Cargo hold monitoring systems • Pre-Arrival Information. • Vessel safe mooring and safe access • Discharge Operation • Cargo discharge checklist	
7. Handle Grain Cargo	• Cargo documentation and shipper's information about the cargo: a. Weight of cargo b. Dimensions c. Securing point details d. Is it palletised e. Are you using ship or shore crane? f. Lashing requirements? g. Weigh bridge certificate • stability criteria for grain at all stages for loading the parcel of grain	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	• Grain cargo receiving preparations. • Vessel safe mooring and safe access. • Loading Meeting: a. loading plan, b. loading sequence and c. final stowage Plan • Cargo inspection prior to loading • Grain Loading operations. • Cargo hold monitoring systems during the voyage. • Information is exchanged with terminal representative at port of discharge, including the discharging plan/sequence. • Grain Discharging operation • Grain Cargo discharge checklist	
8. Handle Dangerous Goods (DG) cargo	• Vessel's cargo documentation and	• Observation • Written assessment

Learning outcomes	content	Suggested assessment methods
	cargo information • IMDG code classification: - Class 1: Explosives - Class 2: Gases - Class 3: Flammable liquids - Class 4: Flammable solids - Class 5: Oxidising agents & organic peroxides - Class 6: Toxins and infectious substances - Class 7: Radioactive material - Class 8: Corrosives - Class 9: Miscellaneous dangerous goods • Preparations to receive the IMDG cargo • Loading plan • Shipboard IMO warning signs - No smoking - Flammable - Do not operate radio communication	• Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	d. Designated smoking area • DG cargo loading precautions • safety and fire rounds during voyage • Discharging operations • IMDG cargo discharge checklist	
9. Handle timber deck cargo	• Timber Cargo documentation and shipper's information about the cargo: a. Weight of cargo b. Dimensions c. Securing point details d. Is it palletized e. Are you using ship or shore crane? f. Lashing requirements? g. Weigh bridge certificate • Stability criteria calculations • Vessel preparations to receive the cargo • Loading operations and	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	stowage • Determining vessel GM using vessel's lifting equipment and prior to departure • lashings inspection during voyage • Risk Assessment and tool box meeting prior to discharge • Information exchange between the vessel and arrival port representatives; • Discharging plan • Cargo handling equipment inspection; • Safe Vessel moored and safe access prior to discharge operations	
10. Perform passenger vessel operations	• Safe means of access to the ship; • Gangway watch; • Muster lists and emergency instructions; • PA system and other means of communicating to passengers; • Lifesaving appliances	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning outcomes	content	Suggested assessment methods
	and personal lifesaving devices; • Passenger screening and boarding passes a. ID checks b. Luggage screening c. Checking in d. Issuing boarding passes e. Attending to passengers with special requirements; • Closing of All watertight openings and side scuttles below the margin line are closed prior to sailing. • Pre-sailing safety announcements. • Call to muster stations and mandatory drills within 24H of sailing including update of Muster list • Drill records • Action to take In case of a death on-board.	

Learning outcomes	content	Suggested assessment methods
11. Inspect and Record Cargo Damages and Defects	• Damage assessment, detailed evidence collection records • Damage report form • Damage reporting to Master, company's P&I club and the stevedoring company • Reporting negligence by stevedores • Role of surveyor in damage survey • documents required by surveyors for damage inspection • Requirement to maintain the ship's seaworthiness during the entire voyage. • Damage survey report • Requirement to Review of the report with crew	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects

- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
• 25 computers with the following software: • Windows/Linux/Macintosh Operating System • Microsoft Office Software • Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection	Cargo handling gear PPE	• Company ISM manual • Register of ships lifting appliances and cargo handling gear • Code of safe working practice (COSWP) • NLS cargo record book • Cargo Securing manual • IMDG code • IBC Code • Grain code • IG Code • Chief officer standing orders • Stability booklet • Hydrostatic particulars • Shippers' information • SOLAS Convention • Bill of lading • Incoterms

WATCHKEEPING PRACTISE

UNIT CODE: 1041 551 19A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Obtained from corresponding unit of competency in Occupational standards

DURATION OF UNIT: 150 HRS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to perform watchkeeping duties. It involves; executing watch taking and handing over, performing watchkeeping duties at anchorage performing watchkeeping duties in port, utilizing meteorological information for safe watchkeeping, performing watchkeeping during loading and discharging of hazardous cargo, performing lookout duties, conducting the vessels in any condition of visibility, transmitting distress signals, performing watchkeeping in coastal and congested waters, conducting the vessels in sight of another, conducting the vessels in restricted visibility, recognizing lights and shapes, recognizing sound and light signals performing watchkeeping while pilot on board and implementing navigational watch procedures.

SUMMARY OF LEARNING OUTCOMES

1. Execute watch taking and handing over;
2. Perform Watchkeeping duties at anchorage;
3. Perform Watchkeeping duties in port;
4. Utilize meteorological information for safe watchkeeping;
5. Perform watchkeeping during loading and discharging of hazardous cargo;
6. Perform lookout duties;
7. Conduct vessels in any condition of visibility;
8. Transmit distress signals;
9. performing watchkeeping in coastal and congested waters;
10. Conduct vessels in sight of another;

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MOMBASA

11. Conduct vessels in restricted visibility
12. Recognize lights, shapes and sound signals
13. Perform watchkeeping while pilot on board
14. Implement navigational watch procedures

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Execute watch taking and handing over.	• depth of water at the berth • ship's draught, the level • time of high and low waters • fastening of the moorings • the arrangement of anchors and the slip of the chain, and other features of mooring; • state of main engines and availability for emergency use; • ongoing works; • nature, amount and disposition of cargo loaded or remaining, or any residue on board; • level of water in bilges and ballast tank; • signals or lights; • Persons onboard: a. crew members b. any other persons on board; • fire-fighting appliances; • special port regulations and communication; • master's standing and special orders;	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• working status of the lines of communication; • Taking over checklist • Handing over as per company procedures. • Status and ability of the relieving officer to carry out his duties; • Operations in continuity. • Handing over checklist.	
2. Perform Watchkeeping duties at anchorage	• Vessel's position; • Monitoring Vessel position at anchor; • Look-out responsibility; • inspection rounds; • meteorological and tidal conditions and state of sea; • Anchor dragging; • state of readiness of the main engines and other machinery; • lights and shapes; • sound signals; • logbook entries and records.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
3. Perform Watchkeeping duties in port.	• Ship to Port communication • safety of the ship and protection of the environment. • moorings and/or anchor chain checks • signals or lights checks; • Safety measures and fire protection regulations; • Arrangements for keeping watch in port • Taking over – handing over watch	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	- Information which the officer being relieved should pass to the relieving officer - Matters on which the relieving officer should satisfy themselves before assuming charge of the watch - Actions on receiving a storm warning or in an emergency - Log book entries	
4. Utilize meteorological information for safe watchkeeping	- wind pressure system and associated wind direction - Beaufort scale interpretation and used - wind speed and direction measurements - cloud types - Requirement to call the Master when in doubt.	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
5. Perform watchkeeping during loading and discharging of hazardous cargo	- Hazardous cargo - safety of the ship and protection of the environment - signals and lights - Safety measures and fire protection regulations; - Personnel requirements when carrying hazardous cargo in bulk - External conditions or circumstances which may imperil the ship - Requirements for special types of ships or cargo - Officer of the watch responsibility	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcome	Content	Suggested Assessment Methods
	- Action in the event of a spillage or fire - Entry into enclosed spaces - Rescue from an enclosed space in an emergency - Requirement to call the Master when in doubt	
6. Perform lookout duties.	- lookout arrangement - Security awareness and vigilance - Conduct fire and security rounds	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
7. Conduct vessels in any condition of visibility.	- Narrow channel - Action taken to avoid collision - Traffic separation scheme - Records and log entries	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
8. Transmit distress signals	- GMDSS and distress call - sound signals - light and shape signals - Ship to Ship Communication	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project

Learning Outcome	Content	Suggested Assessment Methods
		• Practical
9. performing watchkeeping in coastal and congested waters	• Inspection rounds; • Draught and under keel; • State of the weather and sea; • safety precautions and fire protection; • Signals and lights signals; • measures to protect the ship, personnel and cargo in bad weather, or on receiving a storm warning; • Precaution to prevent pollution of the environment; • Assistance from the shore authorities or neighboring ships; • Assistance to ships or persons in distress; • precautions to prevent accidents or damage when propellers are to be turned • Recording keeping and log-book keeping • Call Master when in doubt as per company watchkeeping procedures.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
10. Conduct vessels in sight of another	• Inspection rounds; • signals, sound and/or lights; • Ship-to Shore communication; • Action to take as the stand on vessel in the event the give way vessel does not take proper action • Action as give way vessel • Assistance and response to distress;	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	- Records and log entries - Requirement to call the Master when in doubt.	
11. Conduct vessels in restricted visibility	- Inspection rounds; - Draught and under keel clearance - Safe speed - risk of collision and close quarter situations - Speed reduction - Precautions to prevent accidents or damage in restricted visibility are taken as per company watchkeeping procedures; - Posting additional lookout - Records and Log entries - Requirement to call when in doubt	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
12. Recognize lights, shapes and sound signals	- Lights - Shapes - Lights and shapes for distress - Sound Signals - Action to be taken upon recognizing lights, sound and shapes exhibited by a ship; - records and logs - Requirement to Call Master when in doubt	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
13. perform watchkeeping while pilot on board	- Expected Time of Arrival (ETA) to pilot. - Pilot ladder rigging requirements. - Master/Pilot exchange forms	- Observation - Written assessment - Oral assessment - Third party reports

Learning Outcome	Content	Suggested Assessment Methods
	• Information for the Engine Control Room (ECR); • Manual steering procedures • The pilot flag. • Noting Important navigational marks in the Ship Maneuvering Book correspondent to the chart • Pilotage forms • Authentication of the forms by the Master • Plot vessel's position • Master's instructions with regard to the maneuvering speed.	• Portfolio of evidence • Project • Practical
14. Implement navigational watch procedures	• Officer of the watch responsibility • Principles of watchkeeping • Operational guidance for officers in charge of a navigational watch • Duties of the officer of the watch while at anchor • Log book entries • keep continuous navigation watch when pilot on board • operational guidelines for other duties while on watch; • Operational procedures for areas of high traffic density, restricted visibility and in all hazardous navigational situation • close quarter situation assessment • Time and duration of response for collision avoidance	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	• use the radar and ARPA and limitation awareness: a. ARPA display characteristics b. Performance standards c. Over reliance on ARPA d. Methods of target acquisition • Vessel course and speed during watchkeeping • Fixing ship position at frequent intervals. • Local navigation warnings, and temporary and preliminary notices to mariners • Call Master when in doubt.	
15. Execute Blind Pilotage Techniques	• Pilotage regulations • Items for visual pilotage planning • Items for blind pilotage planning • Route Planning and ETA/ETD • Limiting Danger Line • Methods of track control • Use of edges of land as headmarks/stern marks • ‘No headmark’ procedure • Altering course and monitoring terms • Allowing for a current/tidal stream/leeway when altering course • Monitoring turns • Keeping clear of dangers	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	Blind pilotage preparation and executing techniques	

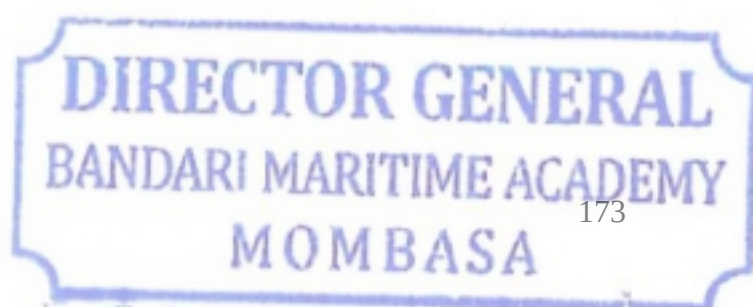
Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 trainees

General Resources	Tools and Equipment	Materials and Supplies
24 computers with the following software: - Windows/Linux/Macintosh Operating System - Microsoft Office Software - Antivirus Software 2 Printers - Printing Papers - External storage media 1 Projector 1 Whiteboard 1 Smartboard/Smart TV (Where applicable) - Assorted whiteboard markers	- Fully equipped seamanship workshop 2 Sextant 2 Azimuth mirror 2 Echo sounders 2 GPS/DGPS 2 Course recorder 2 AIS 2 Speed logs - Access to VDR/SVDR - GMDSS Equipment	- Operating manuals - Company ISM manual - Communication equipment - Flags and shapes - International code of signals - Ships code and decode book - Pyrotechnics - Radar reflector - Ships logbooks - Navigational

General Resources	Tools and Equipment	Materials and Supplies
Internet connection	• Radar • ARPA • ECDIS • Magnetic compass • Gyro Compass • Binoculars • Aldis Lamp • Bell and Gong • Stationery • Chronometer • Calculator • LRIT • BNWAS • Radar and ARPA • Searchlight	charts and publications • Parallel rulers, • pencil, • divider and pair of compasses, • chart correction pen, • rubber,



VOYAGE PLANNING AND OCEAN PASSAGE.

UNIT CODE: 1041 551 21A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Perform Voyage planning and ocean passage.

DURATION OF UNIT: 150 HOURS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to perform voyage planning and ocean passage. It involves; appraising the intended passage, planning and documenting a passage, evaluating a completed passage plan, executing the voyage plan, monitor safe passage of voyage plan, executing weather routing, executing pilotage plan and evaluating the effect of meteorological and climatological processes on passage planning.

SUMMARY OF LEARNING UTCOMES

1. Appraise the Intended Passage.
2. Plan and Document a Passage.
3. Evaluate a Completed Passage Plan.
4. Execute the Voyage Plan
5. Monitor Safe Passage of Voyage Plan.
6. Execute Weather Routing
7. Execute Pilotage Plan
8. Evaluate the Effect Of Meteorological and Climatological Processes On Passage Planning.

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Appraise the Intended Passage.	• Information and Regulations of the intended port • Weather Information	• Observation • Portfolio of Evidence

Learning Outcomes	Content	Suggested Assessment Methods
	- Assessment of Dangers for the intended voyage - Risks assessment for the intended passage - Specific voyage instructions - Recommendations of IMO Conventions, Resolution and Circulars	- Project - Written assessment - Practical assessment - Oral assessment
2. Plan and Document a Passage.	- Tools for Voyage Planning - Nautical Publications and Charts - Berth to Berth Course for the intended passage - Hazards and Marine Protected Areas (MPAs) - Precautionary information for Safe passage - draft passage plan	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
3. Evaluate a Completed Passage Plan.	- Review of Laid Track for The Passage - Adjustment of Waypoints in relation to - Approval of Passage Plan - Signed Passage Plan	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
4. Execute the Voyage Plan	Bridge Team and Engine Team briefing	Observation

Learning Outcomes	Content	Suggested Assessment Methods
	- Condition of Steering Gear and Navigational Equipment; - Expected Time of Arrival (ETA) at critical points - Traffic Condition along the passage - Safe Speed for the route	- Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
5. Monitor Safe Passage of Voyage Plan.	- Ship's Position - Safe passage Monitoring - Marine Environmental Protection Requirement - The Voyage Plan updates	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
6. Execute Weather Routing	- The METAREAs for the planned voyage is set as per the Company navigation procedures - Information of the Focal Point of Contact - meteorological information received via the onboard equipment - Meteorological Information Analyses for voyage planning - Report of The Meteorological Information	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcomes	Content	Suggested Assessment Methods
	Adjustment of Planned Route	
7. Execute Pilotage Plan	- Expected Time of Arrival (ETA) at pilot station - Report to Vessel Traffic Service (VTS) on crossing to the service area - Pilot Station Notication - Pilotage Information for Master - Engine Control Room (ECR) notification - Procedures for Crew to rig pilot ladder and clear anchor - Ship's Speed reduction for pilot boarding - Time of Pilot on Board - Pilot Card	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
8. Evaluate the Effect Of Meteorological and Climatological Processes On Passage Planning.	- Meteorological warnings and data - Meteorological Data Analysis as per company procedures - Report of The Information gathered from the meteorological data - Planned Route review taking considering the gathered meteorological and climatological information	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Suggested Methods of Instruction

- Instructor lead facilitation of practical using active learning strategies.
- Demonstrations

- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainee

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: • Windows/Linux/Macintosh Operating System • Microsoft Office Software • Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	• Sextant • Chronometer • Compass • Calculator	• Company ISM manual • Plotting pad • Pencil • Pen • Eraser • Stick Glue Stick • Violet Ink Pens (Approved by UKHO) • Nautical Charts and Publications • Parallel rule • Compass and dividers

SHIP HANDLING

UNIT CODE: 1041 551 22A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Maintain Ship Stability.

DURATION OF UNIT: 150 HOURS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to perform ship handling. It involves; handling the ship in normal and adverse weather conditions, handling the ship in shallow waters and narrow channels, maneuvering the ship to anchor, perform mooring operations and maneuvering the ship in emergencies.

SUMMARY OF LEARNING OUTCOMES

1. Handle ship in Normal and adverse weather conditions
2. Handle ship in shallow waters and narrow channels
3. Maneuver ship to anchor
4. Perform mooring operations
5. Maneuver the ship in emergencies

Learning Outcome, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Handle ship in Normal and adverse weather conditions	• Handling of ship without wind • Handling of ship without current • Handling of ship with both wind and current	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	- Handling of ship in Heavy and Adverse Weather Conditions: - Restricted visibility - Thunderstorms, - Strong winds - Heavy rainfall - Hail, - Snow - Handling ship in Tropical Revolving Storms (TRS) - Handling of ship during helicopter operations	
2. Handle ship in shallow waters and narrow channels	- Squat - Minimum steerage way - Rudder effect - Bank suction and rejection - Narrow channel checklist - Own ships interaction with other vessels	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment
3. Maneuver ship to anchor	- Anchoring location - water depth, - seabed composition, - weather conditions - ship traffic and density at designated location - Anchor is prepared for letting go - Control of Speed of the vessel while approaching the anchoring position	- Observation - Portfolio of Evidence - Project - Written assessment - Practical assessment - Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	• Testing astern propulsion • Powering and testing bow thruster • Stemming the wind and tide • Walking back the anchor and stopping the vessel at anchoring position • Letting go the anchor	
4. Perform mooring operations	• Mooring method and configuration: a. Mediterranean mooring, b. Baltic mooring, c. Running mooring, d. Standing mooring, e. Spider mooring buoys, f. Single-point or single buoy mooring (SPM/SBM), g. Conventional or multi buoy mooring, h. Ship to Ship Mooring • Approaching the berth/ mooring buoy is executed as per company navigational procedures • Switch on thrusters and make fast the tugs to berth as per company navigational procedures	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

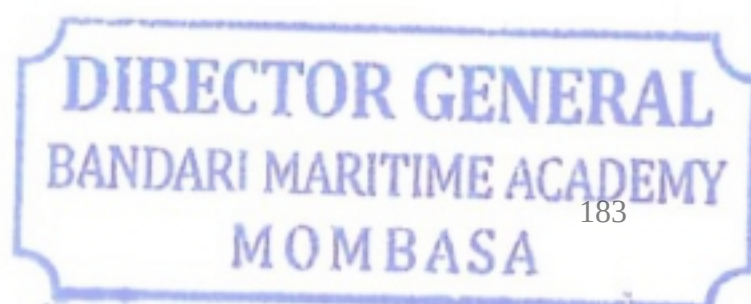
Learning Outcome	Content	Suggested Assessment Methods
	• Send mooring lines to make fast ship as per company navigational procedures • Finish with Engines as per company navigational procedures	
5. Maneuver the ship in emergencies	• Emergency Alarm • MOB maneuvers: a. Williamson turn b. Scharnow turn c. Anderson turn • Posting Lookouts during recovery of MOB • Speed control • Recovery of the MOB • fire alarm • Ship Maneuvers during fire emergency.	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Suggested Methods of Instruction

- Instructor lead facilitation of practical using active learning strategies.
- Demonstrations
- Simulation
- Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainee

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: • Windows/Linux/Macintosh Operating System • Microsoft Office Software • Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection	Fully equipped seamanship workshop Access to a Bridge Simulator	• Company ISM manual • Wheelhouse poster • Maneuvering booklet • Bell book • Anchor log • Deck logbook • Contingency Plans • Bridge procedures guide • Manoeuvring booklet



EMERGENCIES RESPONSE

UNIT CODE: 1041 551 23A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Respond to Emergencies.

DURATION OF UNIT: 120 HOURS

UNIT DESCRIPTION

This unit covers the competencies required by a nautical operations officer to respond to emergencies. It involves; responding to emergencies at sea, responding to emergencies at port, responding to distress signals at sea, performing damage control, implementing the Shipboard Contingency Plan, implementing grounding and beaching procedures, implementing emergency bridge watchkeeping procedures, implementing search and rescue procedures and implementing emergency towing procedures

SUMMARY OF LEARNING OUTCOMES

1. Respond to emergencies at sea.
2. Respond to emergencies at port
3. Respond to distress signals at sea.
4. Perform damage control
5. Implement the Shipboard Contingency Plan
6. Implement grounding and beaching procedures.
7. Implement emergency bridge watchkeeping procedures.
8. Implement search and rescue procedures
9. Implement emergency towing procedures

Learning Outcome, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Respond to emergencies at sea.	• Types of emergencies: a. Fires and Explosions. b. Man Overboard. c. Collisions	• Observation • Written assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	d. Allision e. Stranding/Groundings. f. Severe Weather Conditions. g. Medical Emergencies. h. Hazardous Material Spills. i. Equipment failures—main engine, steering, gyro compass, bridge control j. System failures k. Loss of stability • Alarm response • Preliminary action • Situation Assessment • resources and equipment Mobilization • Emergency handling • Event documentation • Report preparation	• Third party reports • Portfolio of evidence • Project • Practical
2. Respond to emergencies at port	• Emergencies in the Port • Alarm response • Preliminary action • cargo operation stoppage • mustering stevedores • Submission of necessary plans/equipment to shore personnel assisting own ship • Restricted areas onboard • Documentation of chronology of events • incident report	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
3. Respond to distress signals at sea.	• Distress signal • Record of distress signal in GMDSS log book • Requirement for Calling master • Master orders, • Situation reports	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Perform damage control	• Assessment the extent of damage • Damage control plan. • Damage Report	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
5. Implement the Shipboard Contingency Plan	• Shipboard contingency plan. • List and emergency instructions • Emergency response teams duties: a. Command Team b. Emergency Team c. Back Up Emergency Team d. Engine Room Emergency Team • Procedures for safety when approaching heavy weather for hatches, ventilators and security of deck cargo • Emergency drills: a. Fire Drill b. Abandon Ship Drill	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	c. Grounding Drill d. Helicopter Operation Drill e. Collision Drill f. Man overboard Drill g. Engine Room Fire Drill h. Oil Spill Response Drill i. Engine Room flowing drill j. Boat drills k. Chemical spill response drills l. Emergency steering drills m. Enclosed space entry and rescue drills n. Intact stability and damage control drill o. Security Drills and Exercises	
6. Implement grounding and beaching procedures.	• Risk assessment to inform beaching of the vessel • seabed features for beaching the vessel: a. Gently Shelving Beach b. Mud, c. Sand d. Gravel • beaching stern first or head first • beaching the vessel broadside-on or at right angles to the beach • Effect of wind or tide along a shore and the beached ship	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcome	Content	Suggested Assessment Methods
	- Measures to prevent ship driving further ashore - procedures for use of ballast to refloat the vessel. - Sounding of tanks by checking depth of water and if water level is increasing - Sounding round the ship for both grounding and beaching to ascertain depth and nature of bottom	
7. Implement emergency bridge watchkeeping procedures.	- Requirement to call the Master to the bridge in case of emergencies - Action in case of failure of bridge and engine room equipment - Action to respond to emergency situations at sea	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical
8. Implement search and rescue procedures	- Emergency and Distress Signals: - Distress message - Urgency message - Safety message - communication with the distressed craft - information on the position and nature of the distress: - Identification, position, nature and kind of assistance required - Other information	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Learning Outcome	Content	Suggested Assessment Methods
	• contingency planning and master's instructions: a. IAMSAR manual: Action by assisting ships b. Immediate action c. Proceeding to the area of distress d. On board preparation e. Aircraft casualties at sea f. Follow a contingency plan • general arrangements for search and rescue: • Use of the global maritime distress and safety system • assistance provided by authorities and agencies in proximity with to the distress position. • Ships' position and reporting systems • Planning and preparation for a search: a. Categories of distress incidents whether Coastal and ocean b. Communication channels to use whether internal or external c. Co-ordination of search and rescue operations d. Designation of Search and Rescue Mission Coordinators (SMC) and On-scene coordinator (OSC) and their responsibilities	

Learning Outcome	Content	Suggested Assessment Methods
	e. Responsibility of other craft in the coordinated search f. Search patterns taking into account drift, leeway, visibility and type of assistance available g. Drift patterns of disabled vessels with relation to wind and currents h. Rendezvous i. Establishing a datum point before commencing search pattern j. Contingency planning and training • action to conduct a search: a. Proceeding to the area of distress b. On board preparation c. Action by assisting ships d. Approaching the scene, radar search e. Arrival on scene, implement search plan f. Assistance by Search and Rescue (SAR) aircraft g. Homing on radio signals h. Aircraft casualties • action to conduct a rescue • Search and Rescue (SAR) operations terminations procedures:	

Learning Outcome	Content	Suggested Assessment Methods
	- obligations and responsibilities for assistance at sea and the action to be taken to render assistance - Log book entries	
9. Implement emergency towing procedures	- Drawing of fore and aft deck showing emergency towing arrangements - Emergency Towing equipment - Towing Operations - Emergency towing arrangements for tankers of not less than 20,000 DWT	- Observation - Written assessment - Oral assessment - Third party reports - Portfolio of evidence - Project - Practical

Suggested Methods of Instruction

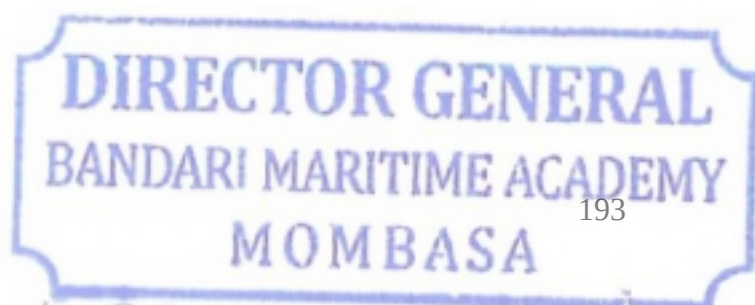
- Instructor lead facilitation of practical using active learning strategies.
- Demonstrations
- Simulation
- Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainee

General Resources	Tools and Equipment	Materials and Supplies
24 computers with the following software:	Fully equipped seamanship workshop	2 Sample Company ISM manual 2 Maneuvering booklet

General Resources	Tools and Equipment	Materials and Supplies
a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector • 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers Internet connection		• 2 Bell book • 2 Anchor log • 2 Deck logbook • 2 Contingency Plans • 2 Bridge procedures guide • 2 Wheelhouse poster • 2 International Aeronautical and Maritime Search and Rescue Manual (IAMSAR) • 2 Convention on the Safety of Life at Sea (SOLAS) • 2 International Regulations for the Prevention of Collisions at Sea (COLREG) • 8 Annual Summary of Notices to Mariners Notice • 2 International Code of Signals (ICS) • 8 International Maritime Organization (IMO) Standard Marine Communication Phrases (SMCP)

General Resources	Tools and Equipment	Materials and Supplies
		• 2 Emergency plan • 2 IAMSAR manual • 8 National emergency response plans • 1 set GMDSS Equipment • 1 set Life Saving Appliances (LSA) • 2 set Firefighting Equipment (FFE) • 1 Decision making support system (DSS) • 2 Damage control plan • 2 Towing booklet • 2 IMDG code • 2 IMO/WHO/ILO Medical first aid guide for use in accidents involving dangerous goods (MFAG)



SHIPPING BUSINESS MANAGEMENT

UNIT CODE: 1041 551 24A

RELATIONSHIP TO OCCUPATIONAL STANDARDS

This unit addresses the Unit of Competency: Manage shipping Business.

DURATION OF UNIT: 120 HOURS

UNIT DESCRIPTION

This unit describes competencies required to manage shipping business. It involves executing shipping business contract, preparing cargo manifest, controlling shipping budget, managing shipping documents, performing cargo connection, handling cargo claims, managing operations resources, performing ship husbandry operations, handling international sales transactions and managing liner ship operations.

SUMMARY OF LEARNING OUTCOMES

1. Perform management duties
2. Implement maritime industry regulations;
3. Implement quality management procedures.
4. Implement ISM procedures;
5. Execute shipping business contract
6. Prepare cargo manifest
7. Control shipping budget
8. Manage shipping documents
9. Perform cargo connection
10. Handle cargo claims

11. Manage operations resources
12. Perform ship husbandry operations
13. Handle international sales transactions
14. Manage liner ship operations

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcomes	Content	Suggested Assessment Methods
1. Perform management duties	• Office Administration and Organizational Methods • Meeting Techniques • Conflict and Change management • Public and Human Relations guidelines • Material Procurement and Stores Control • Customer Relations policies.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
2. Implement maritime industry regulations;	• Shipboard Management • Maritime insurance regulations • shipboard and shore based management national maritime regulations • professional and technical guidelines for operation of ships and maritime facilities • offshore Oil and gas industry guidelines	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
3. Implement quality management procedures.	• Total Quality Management (TQM) • The concept of “Just in Time (JIT)” • ISO integration and interpretation methodologies: a. ISO 9001:2015 b. ISO 14001:2015 c. ISO 50001:2018 d. ISO 45001:2018 • ISO 50001:2018 Energy Management system • cybersecurity ISO standards	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
4. Implement ISM procedures;	• Requirements of the ISM Code • ISO standard requirements in relation to the ISM code • Harmonized "seamless" certification of management systems • Designated Person Ashore (DPA) • ISM compliance, internal audits and the system of external verification • Safety Management Systems related to hazard identification • Risk assessments methodologies -and their role in the ISM Code	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
5. Execute shipping business contract	• Vessel space/vessel • Vessel availability • Cargo readiness • Shipping documents • Calling forward of cargo • Cargo delivery • Contract requirements and specifications • Contract	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
6. Prepare cargo manifest	• Unitised/Bulk Cargo • Loading list/Mate's receipt • Particulars of the manifest • Cargo manifest	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
7. Control shipping budget	• Shipping operations scope • Cost elements • Budget requirements • Budget proposal • Budget proposal approval and implementation • Budget implementation monitoring and evaluation.	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
8. Manage shipping documents	• Shipping documents: a. Proforma invoice b. Commercial invoice	• Observation • Written assessment • Oral assessment

Learning Outcomes	Content	Suggested Assessment Methods
	c. Packing list d. Certificates of origin e. Phytosanitary certificate f. Verified Gross Mass (VGM) Certificate g. Certificate of Conformity h. Shipping instruction i. Bills of lading j. Dangerous goods declaration forms k. Cargo manifest • Screening Shipping documents • Processing Shipping documents • Storage and maintenance of Shipping documents	• Third party reports • Portfolio of evidence • Project • Practical
9. Perform cargo connection	• Receiving and/or accepting cargo from shippers; • Connection of Through-transport cargo • Cargo released to consignees	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
10. Handle cargo claims	• Cargo claims receiving • Cargo claim is investigation • Cargo claim is processing • Cargo claim is settling	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project

Learning Outcomes	Content	Suggested Assessment Methods
		• Practical
11. Manage operations resources	• Operations resources: a. Handling equipment b. Ship/vessel c. Transport vehicles d. Freight container e. Human resource f. Terminal Operating Systems • Resource plan • Resources utilisation • Resource utilisation monitoring • Resource plan reviewing • Resource maintenance	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
12. Perform ship husbandry operations	• Pre-arrival documents a. Crew list b. Passenger list c. Crew effects declaration d. General declaration e. Dangerous goods declaration f. Maritime declaration of health g. Advance notification for waste disposal h. Last ten (10) ports of call • Pre-arrival documents processing • Ship and crew needs:	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical

Learning Outcomes	Content	Suggested Assessment Methods
	a. Ship chandelling b. Crew clearance c. Medical attention d. Bunkering e. Ship maintenance f. Cash on board g. Garbage collection h. Master's cash • Booking Port facilities and services a. Pilotage b. Tugage/towage c. Mooring d. Anchorage e. Berths f. Buoys • Ship clearance a. Inward clearance b. Outward clearance c. Port Health clearance d. Port Authority clearance e. Custom clearance f. Immigration clearance g. Security clearance • Information and reports a. Statement of accounts b. Disbursement account c. Daily operations report d. Arrival/Departure report(s)	

Learning Outcomes	Content	Suggested Assessment Methods
	e. Notice of Readiness (NOR) • Statement of facts/time sheets	
13. Handle international sales transactions	• International Commercial Terms (INCOTERMS) • Sales contract • Cargo shipping • Shipment documents • Cargo delivery	• Observation • Written assessment • Oral assessment • Third party reports • Portfolio of evidence • Project • Practical
14. Manage liner ship operations	• Vessel scheduling • Ship operating schedule • Vessel Maintenance schedule • Shipping cooperation a. Consortiums b. Alliances c. Joint ventures d. Vessel Sharing Agreement (VSA) • Ship management services a. Crewing b. Insurance c. Repair and maintenance d. Ship stores e. Ship operation f. Accounting and procurement g. Bunkering	•

Learning Outcomes	Content	Suggested Assessment Methods
	h. Registration and certification	

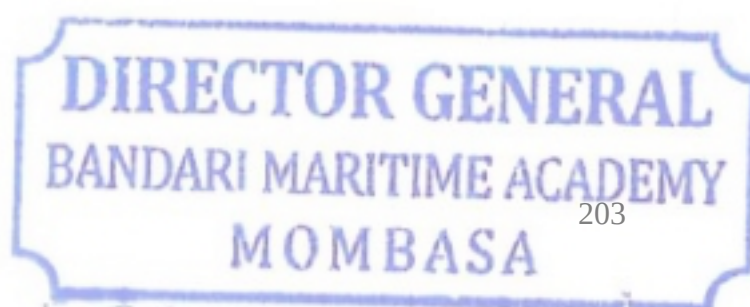
Suggested Methods of Instruction

- Instructor lead facilitation of practical using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 24 Trainee

General Resources	Tools and Equipment	Materials and Supplies
• 24 computers with the following software: a. Windows/Linux/Macintosh Operating System b. Microsoft Office Software c. Antivirus Software • 2 Printers • Printing Papers • External storage media • 1 Projector	• Access to Port Facility	• Sample ship documents • Sample ISM SOPs • Incoterms Chart • Sample Cargo documents

• 1 Whiteboard • 1 Smartboard/Smart TV (Where applicable) • Assorted whiteboard markers • Internet connection		
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ANNEX 1 – STCW TABLE A-II/1

This table shall provide training and assessment guidance during the implementation of this curriculum.

Function: **Navigation at the operational level**

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Plan and conduct a passage and determine position	<i>Celestial navigation</i> Ability to use celestial bodies to determine the ship's position <i>Terrestrial and coastal navigation</i> Ability to determine the ship's position by use of: - landmarks - aids to navigation, including lighthouses, beacons and buoys - dead reckoning, taking into account winds, tides, currents and estimated speed	Examination and assessment of evidence obtained from one or more of the following: - approved in-service experience - approved training ship experience - approved simulator training, where appropriate - approved laboratory equipment training using chart catalogues, charts, nautical publications, radio navigational warnings, sextant, azimuth mirror, electronic navigation equipment,	The information obtained from nautical charts and publications is relevant, interpreted correctly and properly applied. All potential navigational hazards are accurately identified The primary method of fixing the ship's position is the most appropriate to the prevailing circumstances and conditions The position is determined within the limits of acceptable instrument/system errors The reliability of the information obtained from the primary method of position

Column 1	Column 2	Column 3	Column 4
	Thorough knowledge of and ability to use nautical charts, and publications, such as sailing directions, tide tables, notices to mariners, radio navigational warnings and ships' routeing information <i>Electronic systems of position fixing and navigation</i>	echo-sounding equipment, compass	fixing is checked at appropriate intervals Calculations and measurements of navigational information are accurate The charts selected are the largest scale suitable for the area of navigation and charts and publications are corrected in accordance with the latest information available
	Ability to determine the ship's position by use of electronic navigational aids		Performance checks and tests to navigation systems comply with manufacturer's recommendations and good navigational practice
Plan and conduct a passage and determine position (continued)	<i>Echo-sounders</i> Ability to operate the equipment and apply the information correctly		

Column 1	Column 2	Column 3	Column 4
	<i>Compass – magnetic and gyro</i> Knowledge of the principles of magnetic and gyro-compasses Ability to determine errors of the magnetic and gyro-compasses, using celestial and terrestrial means, and to allow for such errors <i>Steering control system</i> Knowledge of steering control systems, operational procedures and change-over from manual to automatic control and vice versa. Adjustment of controls for optimum performance <i>Meteorology</i> Ability to use and interpret information obtained from shipborne meteorological instruments Knowledge of the characteristics of the various weather systems, reporting procedures and recording systems Ability to apply the meteorological information available		Errors in magnetic and gyro-compasses are determined and correctly applied to courses and bearings The selection of the mode of steering is the most suitable for the prevailing weather, sea and traffic conditions and intended manoeuvres Measurements and observations of weather conditions are accurate and appropriate to the passage Meteorological information is correctly interpreted and applied

Column 1	Column 2	Column 3	Column 4
Maintain a safe navigational watch	<i>Watchkeeping</i> Thorough knowledge of the content, application and intent of the International Regulations for Preventing Collisions at Sea, 1972, as amended Thorough knowledge of the Principles to be observed in keeping a navigational watch The use of routeing in accordance with the General Provisions on Ships' Routeing The use of information from navigational equipment for maintaining a safe navigational watch Knowledge of blind pilotage techniques The use of reporting in accordance with the General Principles for Ship Reporting Systems and with VTS procedures	Examination and assessment of evidence obtained from one or more of the following: .1 approved in-service experience; .2 approved training ship experience .3 approved simulator training, where appropriate .4 approved laboratory equipment training	The conduct, handover and relief of the watch conforms with accepted principles and procedures A proper look-out is maintained at all times and in such a way as to conform to accepted principles and procedures Lights, shapes and sound signals conform with the requirements contained in the International Regulations for Preventing Collisions at Sea, 1972, as amended, and are correctly recognized The frequency and extent of monitoring of traffic, the ship and the environment conform with accepted principles and procedures A proper record is maintained of the movements and activities relating to the navigation of the ship Responsibility for the safety of

Column 1	Column 2	Column 3	Column 4
			navigation is clearly defined at all times, including periods when the master is on the bridge and while under pilotage
Maintain a safe navigational watch (continued)	<i>Bridge resource management</i> Knowledge of bridge resource management principles, including: 1. allocation, assignment, and prioritization of resources 2. effective communication 3. assertiveness and leadership 4. obtaining and maintaining situational awareness 5. consideration of team experience	Assessment of evidence obtained from one or more of the following: 1. approved training 2. approved in-service experience 3. approved simulator training	Resources are allocated and assigned as needed in correct priority to perform necessary tasks Communication is clearly and unambiguously given and received Questionable decisions and/or actions result in appropriate challenge and response Effective leadership behaviours are identified Team member(s) share accurate understanding of current and predicted vessel state, navigation path, and external environment
Use of radar and ARPA to maintain safety of navigation Note: Training and assessment in the use	<i>Radar navigation</i> Knowledge of the fundamentals of radar	Assessment of evidence obtained from approved radar simulator and ARPA simulator plus in-	Information obtained from radar and ARPA is correctly interpreted and analysed, taking into account the

Column 1	Column 2	Column 3	Column 4
of ARPA is not required for those who serve exclusively on ships not fitted with ARPA. This limitation shall be reflected in the endorsement issued to the seafarer concerned	and automatic radar plotting aids (ARPA) Ability to operate and to interpret and analyse information obtained from radar, including the following: Performance, including: 1. factors affecting performance and accuracy 2. setting up and maintaining displays 3. detection of misrepresentation of information, false echoes, sea return, etc., racons and SARTs	service experience	limitations of the equipment and prevailing circumstances and conditions
Use of radar and ARPA to maintain safety of navigation <i>(continued)</i> <i>Note:</i> Training and assessment in the use of ARPA is not required for those who serve exclusively on ships not fitted with ARPA. This limitation shall be reflected in the endorsement issued to the seafarer concerned	Use, including: 1. range and bearing; course and speed of other ships; time and distance of closest approach of crossing, meeting overtaking ships 2. identification of critical echoes; detecting course and speed changes of other ships; effect of changes in own		Action taken to avoid a close encounter or collision with other vessels is in accordance with the International Regulations for Preventing Collisions at Sea, 1972, as amended Decisions to amend course and/or speed are both timely and in accordance with accepted navigation practice

Column 1	Column 2	Column 3	Column 4
	ship's course or speed or both 3. application of the International Regulations for Preventing Collisions at Sea, 1972, as amended 4. plotting techniques and relative- and true-motion concepts 5. parallel indexing		Adjustments made to the ship's course and speed maintain safety of navigation Communication is clear, concise and acknowledged at all times in a seamanlike manner Manoeuvring signals are made at the appropriate time and are in accordance with the International Regulations for Preventing Collisions at Sea, 1972, as amended
Use of radar and ARPA to maintain safety of navigation (continued) <i>Note:</i> Training and assessment in the use of ARPA is not required for those who serve exclusively on ships not fitted with ARPA. This limitation shall be reflected in the endorsement issued to the seafarer concerned	Principal types of ARPA, their display characteristics, performance standards and the dangers of over-reliance on ARPA Ability to operate and to interpret and analyse information obtained from ARPA, including: 1. system performance and accuracy, tracking capabilities and limitations, and processing delays		

Column 1	Column 2	Column 3	Column 4
	2. use of operational warnings and system tests 3. methods of target acquisition and their limitations 4. true and relative vectors, graphic representation of target information and danger areas 5. deriving and analysing information, critical echoes, exclusion areas and trial manoeuvres		
Use of ECDIS to maintain the safety of navigation Note: Training and assessment in the use of ECDIS is not required for those who serve exclusively on ships not fitted with ECDIS These limitations shall be reflected in the endorsements issued to the seafarer concerned	<i>Navigation using ECDIS</i> Knowledge of the capability and limitations of ECDIS operations, including: 1. a thorough understanding of Electronic Navigational Chart (ENC) data, data accuracy, presentation rules, display options and other chart data formats 2. the dangers of over-reliance 3. familiarity with the functions of ECDIS required	Examination and assessment of evidence obtained from one or more of the following: .1 approved training ship experience .2 approved ECDIS simulator training	Monitors information on ECDIS in a manner that contributes to safe navigation Information obtained from ECDIS (including radar overlay and/or radar tracking functions, when fitted) is correctly interpreted and analysed, taking into account the limitations of the equipment, all connected sensors (including radar and AIS where interfaced), and prevailing circumstances and conditions

Column 1	Column 2	Column 3	Column 4
	by performance standards in force Proficiency in operation, interpretation, and analysis of information obtained from ECDIS, including: 1. use of functions that are integrated with other navigation systems in various installations, including proper functioning and adjustment to desired settings 2. safe monitoring and adjustment of information, including own position, sea area display, mode and orientation, chart data displayed, route monitoring, user-created information layers, contacts (when interfaced with AIS and/or radar tracking) and radar overlay functions (when interfaced) 3. confirmation of vessel position by alternative means		Safety of navigation is maintained through adjustments made to the ship's course and speed through ECDIS-controlled track-keeping functions (when fitted) Communication is clear, concise and acknowledged at all times in a seamanlike manner

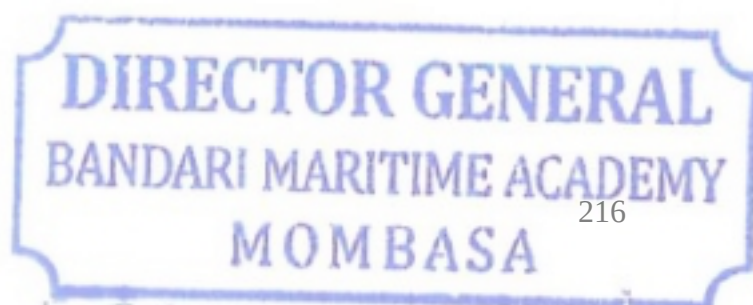
Column 1	Column 2	Column 3	Column 4
	4. efficient use of settings to ensure conformance to operational procedures, including alarm parameters for anti-grounding, proximity to contacts and special areas, completeness of chart data and chart update status, and backup arrangements 5. adjustment of settings and values to suit the present conditions		
Use of ECDIS to maintain the safety of navigation (<i>continued</i>)	6. situational awareness while using ECDIS including safe water and proximity of hazards, set and drift, chart data and scale selection, suitability of route, contact detection and management, and integrity of sensors		
Respond to emergencies	<i>Emergency procedures</i> Precautions for the protection and safety of	Examination and assessment of evidence obtained	The type and scale of the emergency is promptly identified

Column 1	Column 2	Column 3	Column 4
	passengers in emergency situations Initial action to be taken following a collision or a grounding; initial damage assessment and control Appreciation of the procedures to be followed for rescuing persons from the sea, assisting a ship in distress, responding to emergencies which arise in port	from one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved simulator training, where appropriate 4. practical training	Initial actions and, if appropriate, manoeuvring of the ship are in accordance with contingency plans and are appropriate to the urgency of the situation and nature of the emergency
Respond to a distress signal at sea	<i>Search and rescue</i> Knowledge of the contents of the International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual	Examination and assessment of evidence obtained from practical instruction or approved simulator training, where appropriate	The distress or emergency signal is immediately recognized Contingency plans and instructions in standing orders are implemented and complied with

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Use the IMO Standard Marine Communication Phrases and use English in written and oral form	<i>English language</i> Adequate knowledge of the English language to enable the officer to use charts and other nautical publications, to understand meteorological	Examination and assessment of evidence obtained from practical instruction	English language nautical publications and messages relevant to the safety of the ship are correctly

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	information and messages concerning ship's safety and operation, to communicate with other ships, coast stations and VTS centres and to perform the officer's duties also with a multilingual crew, including the ability to use and understand the IMO Standard Marine Communication Phrases (IMO SMCP)		interpreted or drafted Communications are clear and understood
Transmit and receive information by visual signalling	<i>Visual signalling</i> Ability to use the International Code of Signals Ability to transmit and receive, by Morse light, distress signal SOS as specified in Annex IV of the International Regulations for Preventing Collisions at Sea, 1972, as amended, and appendix 1 of the International Code of Signals, and visual signalling of single-letter signals as also specified in the International Code of Signals	Assessment of evidence obtained from practical instruction and/or simulation	Communications within the operator's area of responsibility are consistently successful
Manoeuvre the ship	<i>Ship manoeuvring and handling</i>	Examination and assessment of evidence obtained from	Safe operating limits of ship propulsion, steering and

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	Knowledge of: 1. the effects of deadweight, draught, trim, speed and under-keel clearance on turning circles and stopping distances 2. the effects of wind and current on ship handling 3. manoeuvres and procedures for the rescue of person overboard 4. squat, shallow-water and similar effects 5. proper procedures for anchoring and mooring	one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved simulator training, where appropriate 4. approved training on a manned scale ship model, where appropriate	power systems are not exceeded in normal manoeuvres Adjustments made to the ship's course and speed to maintain safety of navigation



Function: Cargo handling and stowage at the operational level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Monitor the loading, stowage, securing, care during the voyage and the unloading of cargoes	<i>Cargo handling, stowage and securing</i> Knowledge of the effect of cargo, including heavy lifts, on the seaworthiness and stability of the ship Knowledge of safe handling, stowage and securing of cargoes, including dangerous, hazardous and harmful cargoes, and their effect on the safety of life and of the ship Ability to establish and maintain effective communications during loading and unloading	Examination and assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved simulator training, where appropriate	Cargo operations are carried out in accordance with the cargo plan or other documents and established safety rules/regulations, equipment operating instructions and shipboard stowage limitations The handling of dangerous, hazardous and harmful cargoes complies with international regulations and recognized standards and codes of safe practice Communications are clear, understood and consistently successful
Inspect and report defects and damage to cargo spaces, hatch covers and ballast tanks	Knowledge * and ability to explain where to look for damage and defects most commonly encountered due to: 1. loading and unloading operations 2. corrosion 3. severe weather conditions Ability to state which parts of the ship shall	Examination and assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved simulator training,	The inspections are carried out in accordance with laid-down procedures, and defects and damage are detected and properly reported Where no defects or damage are detected, the evidence from testing and examination clearly indicates adequate competence in adhering to procedures and ability to distinguish between normal

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	be inspected each time in order to cover all parts within a given period of time Identify those elements of the ship structure which are critical to the safety of the ship	where appropriate	and defective or damaged parts of the ship
Inspect and report defects and damage to cargo spaces, hatch covers and ballast tanks (continued)	State the causes of corrosion in cargo spaces and ballast tanks and how corrosion can be identified and prevented Knowledge of procedures on how the inspections shall be carried out Ability to explain how to ensure reliable detection of defects and damages Understanding of the purpose of the “enhanced survey programme”		
* It should be understood that deck officers need not be qualified in the survey of ships.			

Function: Controlling the operation of the ship and care for persons on board at the operational level

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Ensure compliance with pollution- prevention requirements	<i>Prevention of pollution of the marine environment and anti-pollution procedures</i> Knowledge of the precautions to be taken to prevent pollution of the marine environment Anti-pollution procedures and all associated equipment Importance of proactive measures to protect the marine environment	Examination and assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved training	Procedures for monitoring shipboard operations and ensuring compliance with MARPOL requirements are fully observed Actions to ensure that a positive environmental reputation is maintained
Maintain seaworthiness of the ship	<i>Ship stability</i> Working knowledge and application of stability, trim and stress tables, diagrams and stress-calculating equipment Understanding of fundamental actions to be taken in the event of partial loss of intact buoyancy Understanding of the fundamentals of watertight integrity <i>Ship construction</i>	Examination and assessment of evidence obtained from one or more of the following: 1. approved in-service experience 2. approved training ship experience 3. approved simulator training, where appropriate 4. approved laboratory equipment training	The stability conditions comply with the IMO intact stability criteria under all conditions of loading Actions to ensure and maintain the watertight integrity of the ship are in accordance with accepted practice

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	General knowledge of the principal structural members of a ship and the proper names for the various parts		
Prevent, control and fight fires on board	<i>Fire prevention and fire-fighting appliances</i> Ability to organize fire drills Knowledge of classes and chemistry of fire Knowledge of fire-fighting systems Knowledge of action to be taken in the event of fire, including fires involving oil systems	Assessment of evidence obtained from approved fire-fighting training and experience as set out in section A-VI/3	The type and scale of the problem is promptly identified and initial actions conform with the emergency procedure and contingency plans for the ship Evacuation, emergency shutdown and isolation procedures are appropriate to the nature of the emergency and are implemented promptly The order of priority and the levels and time-scales of making reports and informing personnel on board are relevant to the nature of the emergency and reflect the urgency of the problem

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
Operate life-saving appliances	<i>Life-saving</i> Ability to organize abandon ship drills and knowledge of the operation of survival craft and rescue boats, their launching appliances and arrangements, and their equipment, including radio life-saving appliances, satellite EPIRBs, SARTs, immersion suits and thermal protective aids	Assessment of evidence obtained from approved training and experience as set out in section A-VI/2 , paragraphs 1 to 4	Actions in responding to abandon ship and survival situations are appropriate to the prevailing circumstances and conditions and comply with accepted safety practices and standards
Apply medical first aid on board ship	<i>Medical aid</i> Practical application of medical guides and advice by radio, including the ability to take effective action based on such knowledge in the case of accidents or illnesses that are likely to occur on board ship	Assessment of evidence obtained from approved training as set out in section A-VI/4, paragraphs 1 to 3	The identification of probable cause, nature and extent of injuries or conditions is prompt and treatment minimizes immediate threat to life
Monitor compliance with legislative requirements	Basic working knowledge of the relevant IMO conventions concerning safety of life at sea, security and protection of the marine environment	Assessment of evidence obtained from examination or approved training	Legislative requirements relating to safety of life at sea, security and protection of the marine environment are correctly identified
Application of leadership and teamworking skills	Working knowledge of shipboard personnel management and training	Assessment of evidence obtained from	The crew are allocated duties and informed of expected standards

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	A knowledge of related international maritime conventions and recommendations, and national legislation Ability to apply task and workload management, including: 1. planning and co-ordination 2. personnel assignment 3. time and resource constraints 4. prioritization Knowledge and ability to apply effective resource management: 1. allocation, assignment, and prioritization of resources 2. effective communication onboard and ashore 3. decisions reflect consideration of team experiences 4. assertiveness and leadership,	one or more of the following: 1. approved training 2. approved in-service experience 3. practical demonstration	of work and behaviour in a manner appropriate to the individuals concerned Training objectives and activities are based on assessment of current competence and capabilities and operational requirements Operations are demonstrated to be in accordance with applicable rules Operations are planned and resources are allocated as needed in correct priority to perform necessary tasks Communication is clearly and unambiguously given and received Effective leadership

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	including motivation 5. obtaining and maintaining situational awareness		behaviours are demonstrated Necessary team member(s) share accurate understanding of current and predicted vessel status and operational status and external environment Decisions are most effective for the situation
Application of leadership and teamworking skills (continued)	Knowledge and ability to apply decision-making techniques: 1. situation and risk assessment 2. identify and consider generated options 3. selecting course of action 4. evaluation of outcome effectiveness		
Contribute to the safety of personnel and ship	Knowledge of personal survival techniques Knowledge of fire prevention and ability to fight and extinguish fires	Assessment of evidence obtained from approved training and experience as set out in section A-VI/1, paragraph 2	Appropriate safety and protective equipment is correctly used Procedures and safe working practices designed

Column 1	Column 2	Column 3	Column 4
Competence	Knowledge, understanding and proficiency	Methods for demonstrating competence	Criteria for evaluating competence
	Knowledge of elementary first aid Knowledge of personal safety and social responsibilities		to safeguard personnel and the ship are observed at all times Procedures designed to safeguard the environment are observed at all times Initial and follow-up action on becoming aware of an emergency conforms with established emergency response procedures

